Engineering Applications of Artificial Intelligence 165 (2026) 113523

Contents lists available at ScienceDirect

Engineering Applications of Artificial Intelligence

journal homepage: www.elsevier.com/locate/engappai

# Application of ant colony optimization with wide-area search concept in path planning for complex large-scale environments

Jianjuan Liu [a,b,*], Yiheng Qian [a,b], Haibo Li [a,b], Miaoxin Ji [a,b], Qiangwei Xu [a,b], Wenzhuo Zhang [a,b]

[a] School of Electrical Engineering, Henan University of Technology, Zhengzhou, 45000, China
[b] Institute of Mechanical and Electrical Equipment and Measurement and Control Technology, Zhengzhou, 45000, China

A R T I C L E I N F O

*Keywords:*
Path planning
Ant colony optimization algorithm
Wide area search strategy
Pheromone diffusion

A B S T R A C T

In complex industrial environments, mobile robot path planning often faces dynamic and expanding maps. Ensuring the robot's safe arrival at its destination is critical when using intelligent algorithms to generate routes. Additionally, considering various aspects of vehicle dynamics, path generation is numerically complex, making traditional path planning algorithms difficult to apply directly. Biological algorithms, with their excellent optimization capabilities, offer new insights into autonomous movement for industrial robots. However, traditional algorithms like Ant Colony Optimization (ACO) suffer from unstable path generation, limiting their real-world applicability. Inspired by biomimetic intelligent vision, this paper proposes a wide-area search strategy to improve the ACO algorithm. The strategy integrates the Probabilistic Roadmap (PRM) method to enhance search efficiency and adaptability. By incorporating additional AI-based strategies to optimize the pheromone mechanism in parallel, the algorithm significantly improves stability. The improved algorithm, named Probabilistic Roadmap-assisted Ant Colony Optimization with Multi-strategy Hybrid and Adaptive Region Search (PR-ACO), demonstrates superior performance over existing methods through both simulation and real-world experiments. PR-ACO exhibits strong adaptability across various map environments through self-learning and consistently generates smoother, shorter paths with reduced computational time.

## 1. Introduction

With the rapid development of automation and e-commerce, the logistics and transportation industry has entered an era of intelligence (Miao et al., 2018). To enhance productivity in industrial processes, the demand for unmanned and intelligent operations in the transportation sector has become increasingly urgent (Butaney et al., 2024). Automated guided vehicles (AGVs), known for their unmanned and high-efficiency advantages, have become essential platforms for logistics handling and exploration tasks (Yaacoub et al., 2020). AGVs are designed to replace humans in hazardous and labor-intensive tasks such as navigating complex terrains, heavy-duty transport, and guided allocation (Li et al., 2019; Tang et al., 2025). They find applications in logistics, warehousing, industrial production, medical assistance (Russell, 2023), mine inspection, agricultural monitoring, military reconnaissance, resource surveying, and rescue operations in harsh environments (Ravankar et al., 2020).

Intelligent transport vehicles are essentially a type of mobile robot (Kanna et al., 2024). The smooth operation of mobile robots relies heavily on their intelligent systems (Liu et al., 2025; Cheng et al., n.d.). As a key component of these systems, path planning is fundamental to ensuring efficient and safe movement, making path planning algorithms a major research focus (Tan et al., 2023). However, in practical applications such as warehousing, logistics, or exploration and rescue, AGVs often face environments that are expansive, complex, and frequently changing (Corominas et al., 2023). Obstacles in the environment may shift dynamically, requiring robots to perform local obstacle avoidance (Lou et al., 2025). When deviations from the predetermined route occur, global path planning algorithms must promptly update to generate new viable paths (Luo et al., 2019). The challenge lies in achieving real-time and reliable path planning under ever-changing conditions (Yu, 2019).

Traditional single-path planning algorithms, such as A*, Ant Colony Optimization (ACO), Artificial Potential Field (APF), Probabilistic Roadmap (PRM), and Particle Swarm Optimization (PSO), each have

* Corresponding author. School of Electrical Engineering, Henan University of Technology, Zhengzhou, 45000, China.
*E-mail address:* liujuanhaut@haut.edu.cn (J. Liu).

https://doi.org/10.1016/j.engappai.2025.113523
Received 19 January 2025; Received in revised form 12 October 2025; Accepted 8 December 2025
Available online 12 December 2025
0952-1976/

their strengths in search capabilities. However, they also exhibit certain limitations when used individually (Yi et al., 2020). A common approach to enhance their search efficiency is to integrate multiple algorithms (Liang et al., 2025). While these hybrid methods often perform well in fixed and relatively small-scale maps, challenges arise in dynamic environments like warehousing and logistics, where maps expand and conditions change frequently (Mao et al., 2025).

Under such complexities, existing algorithms face limited search scope, leading to prolonged computation times and lack of global path optimization (Wu et al., 2023). These issues often result in paths falling into local optima or exhibiting poor smoothness. Consequently, these limitations hinder the adaptability of the planned paths to practical operational needs, particularly in large-scale environments (Liu et al., 2022). Addressing these challenges calls for the development of a new and more robust search algorithm.

The Ant Colony Optimization (ACO) algorithm, known for its superior evolutionary ability and stability, can effectively optimize search performance by improving its heuristic formula and pheromone iteration process (Song et al., 2024). Additionally, ACO's solid iterative model allows for integration with various algorithmic strategies, demonstrating its excellent adaptability (Cao et al., 2019). The Probabilistic Roadmap (PRM) algorithm, with its high search efficiency and broad exploration range, offers significant value (Guo et al., 2022).

In recent years, scholars both at home and abroad have proposed a variety of insightful approaches regarding the application and improvement of intelligent algorithms in global path planning. However, many of these improvements tend to overemphasize a single aspect, leading to persistent limitations in existing enhanced path planning algorithms. Overall, current research efforts have mainly focused on the following aspects of path planning:

1. Path length is a key metric for evaluating the quality of a planned route, and some researchers have focused on reducing the length of the generated paths. Tan et al. integrated the ACO with the particle swarm optimization (PSO) algorithm to optimize the parameters of the ACO for spot welding robots, achieving shorter path lengths and faster convergence (Tan et al. ,2023). In robotics, Cai et al. applied the firefly algorithm and heuristic functions to improve ACO performance for coordinated path planning of multi-mobile robots in 2D and 3D environments, demonstrating reduced planning times and shorter paths (Cai et al. , 2024).
2. Path smoothness essentially determines the feasibility of a planned route in real-world environments. Excessively tortuous paths are often impractical for actual use, and many researchers have proposed various improvements to address this issue. Fang et al. fused ACO with an improved Q-Learning algorithm for search and rescue robots, achieving smoother paths using Bessel curves while balancing exploration and exploitation (Fang et al., 2024). Han et al. developed a 3D path planning approach for coordinate measuring machines, integrating collision detection and artificial potential field strategies for safer paths (Han et al., 2017). Dai et al. proposed a novel curvature suppression operator to refine the heuristic formula of the ant colony algorithm, enhancing path smoothness while strengthening the algorithm's convergence capability (Gong et al., 2018).
3. Optimization of algorithm convergence time is also an important consideration. If the algorithm takes too long to generate a path, it becomes unfavourable for practical applications. To address this, several researchers have proposed improvements targeting convergence efficiency. Cheng et al. improved the ACO by employing an analytic hierarchy process (AHP) to dynamically optimize paths in intelligent transportation systems, resulting in reduced travel time and improved accuracy (Cheng et al. ,2023). Wu et al. proposed a game-theoretic multi-ant colony optimization algorithm with a hierarchical temporal memory model, enhancing convergence and accuracy in large-scale problems (Wu et al., 2024). Wang et al. combined RRT* and ACO for AGV path planning in parking systems, leveraging a dynamic pheromone strategy to improve search speed and robustness in large-scale environments (Wang et al., 2024). Fu et al. proposed a bidirectional artificial potential field-based ant colony optimization (BAPFACO) algorithm to improve mobile robot path planning, addressing challenges such as slow convergence speed, susceptibility to local optima, and trap avoidance, by incorporating bidirectional potential fields, an adaptive heuristic function (Fu et al., 2024). The aforementioned studies have effectively addressed specific aspects of the path planning problem.

However, more practical and comprehensive challenges—such as path planning in large-scale map environments—remain insufficiently resolved. As a result, some researchers have proposed more novel and holistic improvement strategies, aiming to tackle the path planning problem in a broader and more practical context. For example: Lee et al. proposed a strategic artificial potential field (SAPF)-based Q-learning algorithm for path planning in battlefield environments, focusing on utilizing obstacles for concealment and cover (Lee and Seo, 2024). This type of map coverage learning improvement offers a new perspective for addressing the adaptability of algorithms to different map environments. Meanwhile, some researchers have also achieved promising results by introducing various strategies to enhance the intelligence of ant colony optimization algorithms. For example: Shi et al. combined simulated annealing and ACO for cleaning robots, achieving complete area coverage with minimal repetition (Shi et al., 2024). For unmanned surface vehicles (USVs), Xia et al. (2019) introduced a quantum ACO algorithm that optimized path length, energy consumption, and safety, significantly improving efficiency and convergence (Xia et al., 2019). Jiao et al. employed a polymorphic ACO for smart wheelchair path planning, resolving deadlock issues and achieving high optimization efficiency (Jiao et al., 2018). Similarly, Miao et al. proposed an adaptive ACO that enhanced safety and real-time performance for indoor mobile robots (Miao et al., 2021). In industrial and military applications, Liu and Liu introduced a compound prediction mechanism within ACO for mobile robot path planning, achieving faster convergence and better optimization accuracy (Liu and Liu,2023). In addition, Gao et al. (2020) improved the search strategy of the ant colony algorithm by introducing the heuristic formula of the A* algorithm and optimized its convergence capability based on a pheromone diffusion strategy (Gao et al., 2020). However, the aforementioned improvement approaches still have notable limitations. Most researchers have not fully considered the comprehensive application of path planning algorithms in complex and dynamic physical environments. Nevertheless, their work offers valuable inspiration and evolutionary perspectives for the research conducted in this paper.

To address practical challenges in path planning, this paper proposes a novel Probabilistic Roadmap-assisted Ant Colony Optimization algorithm with Multi-strategy Hybrid and Adaptive Region Search (PR-ACO). This algorithm combines the global search capability of the Probabilistic Roadmap (PRM) with the robustness of Ant Colony Optimization (ACO), aiming to enhance path planning performance in large-scale, complex, and dynamic environments. Compared with conventional ACO-based approaches, PR-ACO demonstrates superior performance across diverse map scenarios by achieving faster convergence and identifying higher-quality paths. The paths generated by PR-ACO exhibit stronger global optimality, fewer turns, and shorter overall distances, even in maps of varying sizes and complexities. The core innovations of the proposed PR-ACO algorithm for improving path planning in large, dynamic, and complex environments are summarized as follows:

1. Wide Area Search Strategy with adaptive radius: Traditional ACO algorithms using fixed eight-directional search often fail to efficiently explore large-scale environments. Inspired by the ideas of Particle Swarm Optimization and PRM, this study introduces a dynamic radius wide-area search strategy with recognition capability

for the first time. A novel dynamic weighting framework leverages the routing expertise of elite ants during node selection. By analysing the complexity of the environment, the improved algorithm adaptively expands the search radius, enhancing both spatial coverage and global search efficiency without sacrificing speed. The integration of ant iteration data with elite pheromone counting accelerates convergence while maintaining solution diversity.
2. Extended heuristic function for improved discrimination: Conventional ACO heuristic functions often lack sufficient discriminative power. To address this, an extended heuristic formulation is proposed based on the wide-area search strategy. Different heuristic weights are assigned to candidate nodes in inner and outer regions, significantly improving the ants' ability to distinguish navigational choices and increasing the probability of escaping local optima.
3. New Expansion Strategy Supporting the Hierarchical Decision-Making Elite Ant Optimization Strategy for overcoming premature convergence: Most ACO algorithms employing elite ant systems lack sufficient refinement of the elite strategy, making them ineffective in addressing the issue of convergence to suboptimal solutions. In this study, we propose a new hierarchical decision-making elite ant optimization strategy, which replaces the traditional static elite mechanism with a three-tier optimization protocol based on route performance. The retained elite ant paths are categorized into three types: elite routes, optimizable routes, and non-optimizable routes. Each category is processed with different strategies such as retention, replacement, or elimination. Moreover, the feedback from this layered elite mechanism is used to build a cooperative adaptive pheromone adjustment system, which dynamically tunes the pheromone weighting parameters based on the elite ants' performance. This cooperative and adaptive layered elite strategy not only improves stability during optimization but also effectively mitigates premature convergence and helps avoid suboptimal solutions.
4. Pheromone Topological Diffusion Strategy for global reinforcement: Traditional ACO approaches often lack comprehensive pheromone coverage and make limited use of elite routes. To address this, a pheromone topological diffusion mechanism is proposed, which diffuses pheromone information from improved elite ants toward potential high-quality regions. Unlike conventional localized reinforcement, this mechanism forms a broad-spectrum chemical guidance network, enhancing the algorithm's global self-optimization ability.

Section 2 introduces the fundamental principles of the Ant Colony Optimization (ACO) algorithm and the grid-based map representation method. This section provides the theoretical foundation for the subsequent algorithm design and experimental analysis. Section 3 presents the key innovations of the proposed PR-ACO algorithm, focusing on improvements in the search scope and heuristic function. Specifically, it elaborates on the wide-area search strategy, the dynamic weighting parameter selection mechanism, and the enhancements to the heuristic formulation. Section 4 discusses the pheromone update innovations in PR-ACO. It includes an improved elite ant decision-making mechanism, a dynamic parameter adjustment framework, and a proposed pheromone weighting mechanism for better adaptability in dynamic environments. Section 5 summarizes the experimental results, including both simulation and physical experiments. The results demonstrate that the proposed PR-ACO algorithm significantly outperforms traditional ACO and its enhanced variants under various environmental settings, showing marked improvements in performance and robustness. Finally, Section 6 concludes the paper by summarizing the main findings and outlining potential directions for future research.

In the final experiments, both simulated and real-world, the results demonstrate that the improved PR-ACO algorithm outperforms the traditional ACO algorithm in various environments, showing significant optimization in algorithm performance.

## 2. Introduction to research methods

Therefore, an introduction to the parameters and principles of the ACO algorithm is provided first.

### 2.1. Introduction to ant colony optimization

The Ant Colony Optimization (ACO) algorithm, inspired by the foraging behavior of ants, is widely used in optimization problems (Wang et al., 2024). When ants search for food, they deposit pheromone on the paths they travel, serving as a guide for other ants (Liu et al., 2017). Over time, the concentration of pheromone on each path is determined by factors such as the number of ants that traverse the path and its length. Paths with higher pheromone concentrations attract more ants, creating a positive feedback loop. Conversely, longer paths see fewer ants and lower pheromone accumulation, gradually reducing their attractiveness. Through repeated iterations, pheromone levels on shorter and more optimal paths increase, enabling the colony to converge on the best solution. The mathematical representation of the traditional ACO path-selection process is typically expressed as:

$$P_{ij}^{k}(t) = \begin{cases} \dfrac{\tau_j{}^{\alpha}(t)\cdot\eta_{ij}^{\beta}(t)}{\sum\limits_{s\in allow_k}\tau_j{}^{\alpha}(t)\cdot\eta_{ij}^{\beta}(t)}, & s \in allow_k \\ 0, & s \notin allow_k \end{cases} \tag{1}$$

$$\begin{cases} \eta_{ij}(t) = \dfrac{1}{d_{ij}} \\ d_{ij} = \sqrt{\left(x_i - x_j\right)^2 + \left(y_i - y_j\right)^2} \end{cases} \tag{2}$$

In the Ant Colony Optimization (ACO) algorithm, the pheromone concentration on path grid $j$ is denoted as $\tau_j(t)$, while $\eta_{ij}(t)$ refers to the heuristic information factor associated with the same path. K is the number of iterations. The parameter α determines the influence of pheromone concentration on an ant's path selection, and β controls the weight of heuristic information in the decision-making process. The distance between node $i$ and node $j$ is represented as $d_{ij}$,with $(x_i, y_i)$ and $(x_j, y_j)$ being the coordinates of grid $i$ and grid $j$, respectively. The set $allow_k$ defines the available grids an ant can move to from grid $i$, excluding obstacle grids or those marked as tabu (representing the traversable free grids around $i$). The pheromone update mechanism is expressed using Equation (3).

$$\begin{cases} \tau_i(t+1) = (1-\rho)\tau_i(t) + \Delta\tau_i(t) \\ \triangle\tau_i(t) = \sum\limits_{k=1}^{m}\triangle\tau_i(t)^k \end{cases} \tag{3}$$

Here, $m$ represents the number of ants. $\rho$ is the evaporation rate of pheromone. $\tau_i(t)$ denotes the pheromone concentration at grid $i$ during the t-th iteration. The $\triangle\tau_i(t)^k$ represents the increment of pheromone left by ant $k$ at grid $i$. In this paper, a cyclic system model is adopted, and $\triangle\tau_i(t)^k$ is defined as follows:

$$\Delta\tau_i(t)^k = \begin{cases} \dfrac{Q_1}{L_k(t)}, & \text{if ant } k \text{ found food and passed grid } g_i \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

$Q_1$ is a constant. $L_k(t)$ is the length of the path searched by ant k

### 2.2. Grid method

This study adopts a grid-based approach for robot environment planning, dividing the workspace into an $MM*MM$ grid composed of square cells (Gao et al., 2023). Obstacles are represented as black cells, while free space is denoted by white cells. To ensure collision-free

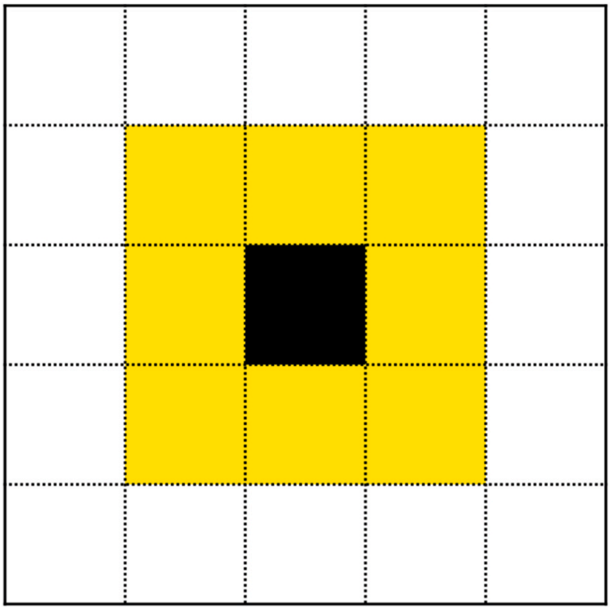


**Fig. 1.** In this context, the black grid represents the current position of the ant colony, while the yellow grid represents the area covered by a single search of the ant. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

movement, obstacles are expanded to occupy entire cells. The grid layout uses a two-dimensional coordinate system for calibration, with each cell assigned a unique number. In an *MM*MM* map, the grid coordinates $(x, y)$ corresponding to a cell labeled $R$ can be expressed as:

$$\begin{cases} x = \begin{cases} mod(R, MM) - 0.5 \\ MM + mod(R, MM) - 0.5 \end{cases} \quad if\ mod(R, MM)! = 0 \\ y = MM + 0.5 - ceil\left(\frac{R}{MM}\right) \end{cases} \tag{5}$$

## 3. Detailed explanation of the improvement method

This section primarily introduces the improved wide-area search strategy and the enhancements made to the heuristic formula, including comparative analysis of these improvements and the introduction of dynamic weight parameters.

### 3.1. Wide Area Search Strategy

Traditional eight-directional search refers to searching the surrounding grids in eight directions from the current path point. These eight directions include up, down, left, right, and the four diagonal directions (top-left, top-right, bottom-left, and bottom-right), as shown in Fig 1 (Chen et al., 2022). In fact, this approach results in a uniform search view. Although it may perform well in some simulation environments, as the map size increases, this strategy becomes less effective in real-world scenarios (Feng et al., 2024). Traditional search strategies struggle with handling local optima in large maps. Additionally, in wider maps, these algorithms may cause some route optimization methods to fail. There is also the issue that the paths found are often not directly useable, with significant detours, and the routes are not the shortest.

This paper first introduces a novel search strategy that analyzes the size and complexity of the constructed map to implement a dynamic wide-area search strategy. By employing adaptive radius selection and a newly designed randomized selection approach, the wide-area search strategy enables the robot to better utilize available information and mimic human navigation behavior—balancing both local and global search processes. Drawing inspiration from the random selection strategy in the Probabilistic Roadmap (PRM) algorithm, this wide-area search strategy enhances both globality and randomness. The algorithm extends the range of grid points considered for selection from the current point. Specifically, within a defined radius, a certain number of reachable grid points are randomly selected, as shown in Fig 2. These candidate points are then added to the set of feasible points for subsequent evaluations, and the ant colony algorithm proceeds to determine the next point based on probabilistic selection.

This approach significantly broadens the scope of each search iteration, improving the algorithm's global search capabilities. By increasing the breadth of each individual search, the number of evaluations required per search iteration is effectively reduced, leading to a noticeable acceleration in search speed.

The wide-area search strategy significantly improves the global performance of the algorithm, enhancing both its efficiency and overall effectiveness. However, in practical applications, attention must be given to the appropriate selection of the search radius in different environments. This paper introduces a dynamic search radius formula, specifically designed to adapt to varying environmental conditions. The dynamic formula is expressed as follows:

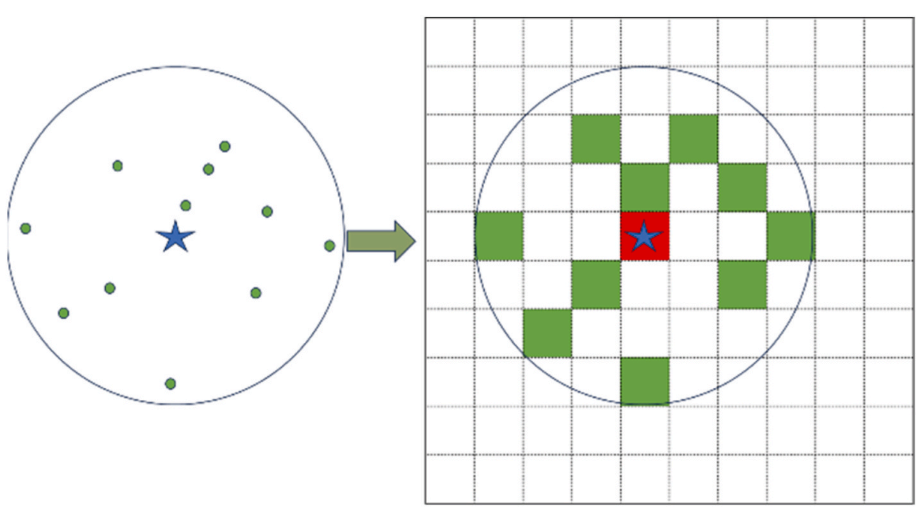


**Fig. 2.** The wide-area search strategy selects viable grids by randomly choosing cells within the two-dimensional search area. The green grid represents the selected viable grid. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

$$r = \max\left(r_{\min}, \frac{1}{\varepsilon} \cdot \sqrt{a^2 + b^2} \cdot (1 - k \cdot p)\right) \tag{6}$$

$$p = \frac{b_o \times l^2}{a \times b} \tag{7}$$

This $r$ represents the generated search radius. $a$ and $b$ are determined by the size of the map currently recognized by the robot. $a$ represents the length of the experimental map, while $b$ represents its width. There $r_{\min}$ is the minimum search radius, ensuring that the radius does not become excessively small. $k$ is the scaling factor used to adjust the overall size of is the scaling factor used to adjust the overall size of $p$. Since the grid method is used in this study to analyse the map, $b_o$ represents the number of grid cells corresponding to obstacles. Here, $l$ denotes the side length of each grid cell, which can be freely chosen and is typically set to 1 in the simulation experiments. Parameters such as $a$, $b$, and $l$ together form $p$, which represents the proportion of the map area occupied by obstacles. The parameter $p$ partially influences whether the search radius $r$ is chosen conservatively or aggressively.

By employing the dynamic radius selection, the algorithm's adaptability and optimization capabilities across diverse environments are significantly enhanced. This allows the algorithm to dynamically adjust its search parameters in real-time when operating in large maps, enabling faster identification of viable paths.

Moreover, the improved algorithm can tailor its search strategy based on the specific characteristics of the environment. For instance, in open maps with fewer obstacles, the algorithm adopts a more aggressive strategy by selecting larger search areas. Conversely, in maps with a high density of obstacles, the algorithm opts for a more conservative approach, utilizing a smaller search radius to navigate effectively. This adaptability ensures robust performance in both sparse and complex environments, demonstrating substantial optimization in the algorithm's search capabilities.

### 3.2. Dynamic selection weight parameters

The improved search algorithm demonstrates strong adaptability in global search scenarios. The proposed enhancements also exhibit exceptional global search performance during optimization, enabling the algorithm to maintain the iterative optimization capabilities of the ACO algorithm even in the later stages of iteration. This is primarily due to the high degree of randomness inherent in the traditional Probabilistic Roadmap (PRM) algorithm (Huang et al., 2024), which generates completely independent random search points during each iteration. While this randomness enhances the global exploration capabilities of the ACO algorithm, it also undermines its convergence performance. Consequently, the improved algorithm may experience oscillations in the later stages, struggling to converge to an optimal path.

To address this issue, this paper proposes a novel random selection weighting function that incorporates elite ants and pheromone information to optimize the original selection mechanism. This adjustment ensures that the improved algorithm maintains the global exploration capabilities of ACO during the early search phases while significantly enhancing its convergence efficiency in the later stages. The modified random selection formula and dynamic weighting parameters are defined as follows:

$$P_j = \frac{\lambda_j}{\sum\limits_{s \in allow_k} \lambda_j} \tag{8}$$

$$\lambda_j = \left[\frac{1}{1 + e^{-(t - t_{\text{thresh}})}} \cdot \left(\frac{n_j^{0.5}}{a_{\max}}\right) + \left(1 - \frac{1}{1 + e^{-(t - t_{\text{thresh}})}}\right) \cdot \left(\frac{a_{\min}}{n_j^{0.5}}\right)\right] \cdot k_1 + k_2 \cdot \tau_j \tag{9}$$

Here, $P_j$ refers to the probability of selecting this grid as the next removable random grid $j$. Where $\lambda_{ij}$ refers to the weight of the grid $j$ selected as a randomly available point. $t$ represents the current iteration number. $t_{thresh}$ indicates the threshold iteration number for determining behaviour. $n_j$ represents the number of times the current grid has been selected as an elite point. $a_{\max}$ and $a_{\min}$ are the maximum and minimum scaling values, respectively, which determine the weight size of the generated parameter. $k_1$ and $k_2$ are adjustment factor parameters. To prevent any single factor from dominating the random selection of feasible points, $k_1$ and $k_2$ are used to proportionally adjust individual parameter weights, resulting in a more balanced probability distribution for point selection. $\tau_j$ refers to the pheromone concentration of the grid. In random point selection, relying solely on pheromone concentration results in insufficient distinction, making the algorithm prone to local optima in the early stages and lacking optimization capabilities in the later stages. Therefore, two input criteria are used to determine the generation weight of random points. First, an elite point parameter system is constructed based on the elite ant system. When a grid is randomly selected as an elite point, its elite weight $n_j$.Second, the current iteration number $t$ and a predefined iteration threshold $t_{thresh}$ are introduced. When $t < t_{thresh}$,the function shows a monotonically decreasing trend, with the weight decreasing $\lambda_j$ as the number of $n_j$ increases. Conversely, when $t > t_{thresh}$, the function exhibits a monotonically increasing trend, with the weight increasing as the number of elite points rises.

This dual-criteria approach allows the algorithm to adapt dynamically across different stages of iteration. In the early stages, the focus is on maintaining global exploration capabilities by prioritizing diversity in random point selection. In the later stages, the emphasis shifts to enhancing convergence by reinforcing the influence of elite points and pheromone concentration. This balance ensures that the algorithm avoids premature convergence while maintaining strong optimization capabilities throughout the iterative process.

### 3.3. Improvement of heuristic formula

The heuristic formula of the Ant Colony Optimization (ACO) algorithm largely determines its local search performance (Sui et al., 2023). During the algorithm's operation, the local search capability plays a critical role in the search efficiency of ACO. However, the traditional ACO heuristic formula has a relatively simplistic way of identifying local costs. Especially during the early stages of the search, when the global pheromone concentration is nearly uniform, the local search capability almost entirely determines the upper limit of the ACO algorithm's pathfinding performance. A well-designed heuristic formula can significantly accelerate the discovery of high-quality paths.

In traditional ACO algorithms, the heuristic formula's reference parameters are overly simplistic. Since ants primarily rely on $d_{ij}$ to identify viable points, as shown in Equation (2), the differences between neighboring grids in traditional formulas are relatively small. This lack of heuristic information about the target position is a major reason for the low search efficiency of ants. To address the problem of prolonged blind searching during iterative optimization, some researchers have proposed using the A* heuristic formula to optimize the traditional ACO search (Hu et al., 2025). Thus, a more commonly used heuristic formula can be expressed as:

$$\eta_{ij}(t) = \frac{1}{d\left(g_i, g_j\right) + d\left(g_j, g_{\text{target}}\right)} \tag{10}$$

Here, $d\left(g_i, g_j\right)$ refers to the distance between the current grid $g_i$ and an adjacent viable grid $g_j$, while $d\left(g_j, g_{\text{target}}\right)$ refers to the distance from the adjacent viable grid to the destination. Together, these values form the local search cost used in the evaluation. This heuristic formula enhances the ants' search capability by incorporating heuristic information about the distance from viable points to the target. However, in practical use, during the early stages of pathfinding, when ants are still

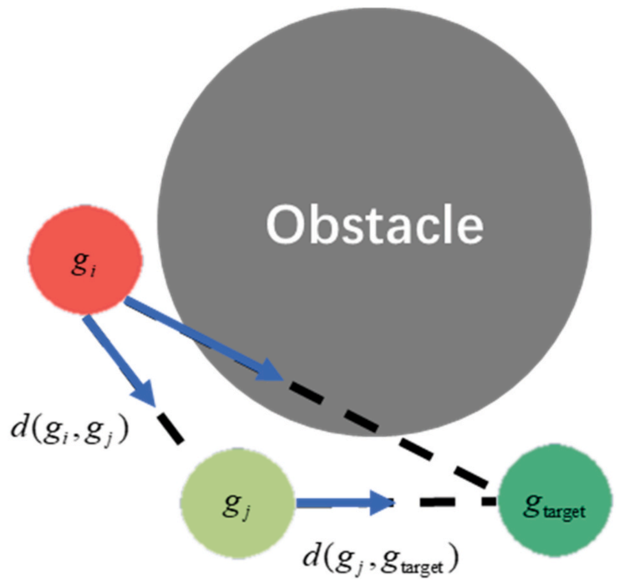



**Fig. 3.** As shown in the figure, $d(g_i, g_j)$ refers to the distance between the current grid $g_i$ and an adjacent viable grid $g_j$. $d(g_j, g_{\text{target}})$ refers to the distance from the adjacent viable grid $g_j$ to the destination $g_{\text{target}}$.

far from the destination, $d(g_j, g_{\text{target}})$ becomes large, and the local distance $d(g_i, g_j)$ becomes much smaller than the heuristic value. This is disadvantageous for the initial search.

On the other hand, due to the traditional eight-direction search method, the distances between viable adjacent points are similar, and the values of $d(g_i, g_j)$ are almost identical. As shown in Fig 3, the differences in the overall distance $d(g_j, g_{\text{target}})$ are not significant, and the heuristic information is weak. This makes the traditional heuristic formula's blind search problem more pronounced in large maps. Furthermore, if directly applied to the practical wide-area search strategy, some problems become even more difficult to solve. For instance, in certain situations, ants are more likely to enter "trap zones," as shown in Fig 5, where the uneven distribution of search weights makes it harder to escape

In order to strengthen the heuristic information and address the issues of blind searching and trap zones, this paper proposes a new heuristic formula, combined with the wide-area dynamic search strategy, as follows:

$$\eta_{ij}(t) = \frac{1}{\frac{d(g_i, g_j)}{r} + \left(1 + \frac{C_1}{d(g_i, g_j)}\right) * d(g_j, g_{\text{target}}) + C} \tag{11}$$

Here, $r$ represents the search radius, $C$ and $C_1$ are weight adjustment parameters. In experiments, variations in the initial selection of pheromone concentration and update parameters may cause an imbalance between $\eta_{ij}(t)$ and $\tau_i(t)$. To address this issue, $C$ is used to adjust the proportion of $\eta_{ij}(t)$, while $C$ and $C_1$ serve similar roles in preventing the computed parameters from becoming excessively large or small. $d(g_i, g_j)$ represents the distance between the current grid and the adjacent viable grid in the improved wide-area search, as shown in Fig 4(b), providing greater differentiation. It can be seen that under the traditional heuristic formula, ants tend to fall into trap zones Fig 5 due to uneven search weight distribution. By using $d(g_i, g_j)$ and $r$, a dynamic weight parameter is created to improve the distance difference between the ant's current position and the viable grid. This ensures ants have a more balanced cost choice between inner and outer grids. Additionally, the dynamic cost weight built with dd clarifies the heuristic information for the distance from the viable grid $g_j$ to the destination $g_{\text{target}}$, allowing ants to prioritize grids closer to the target. This reduces redundant searches and improves efficiency. Ants can now search more clearly and escape trap zones, boosting overall search performance.

## 4. Pheromone update strategy and elite strategy construction

The ACO algorithm, as a biologically inspired algorithm based on learning and iterative updates, relies heavily on the pheromone update mechanism for its key strategy. Ants are closely connected through the pheromone update process, and it is through the retention of

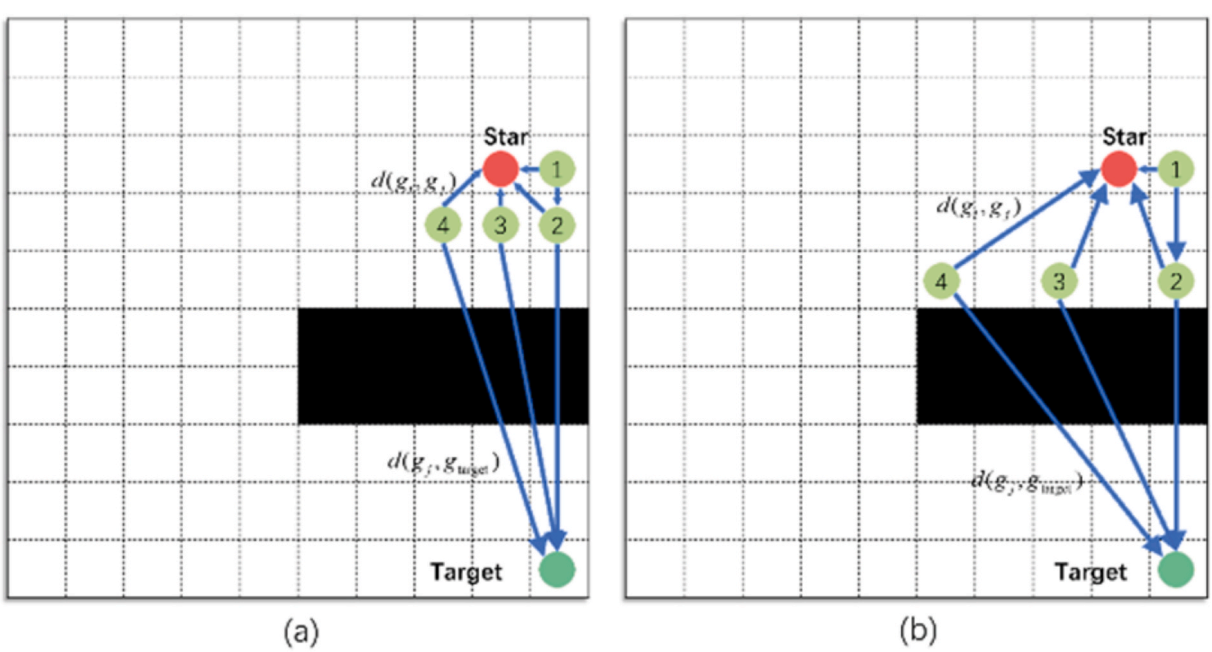



**Fig. 4.** (a) Shows the search situation using the traditional eight-directional search with the conventional heuristic formula. (b) Shows some possible search scenarios using the wide-area search theory proposed in this paper with the traditional heuristic formula.

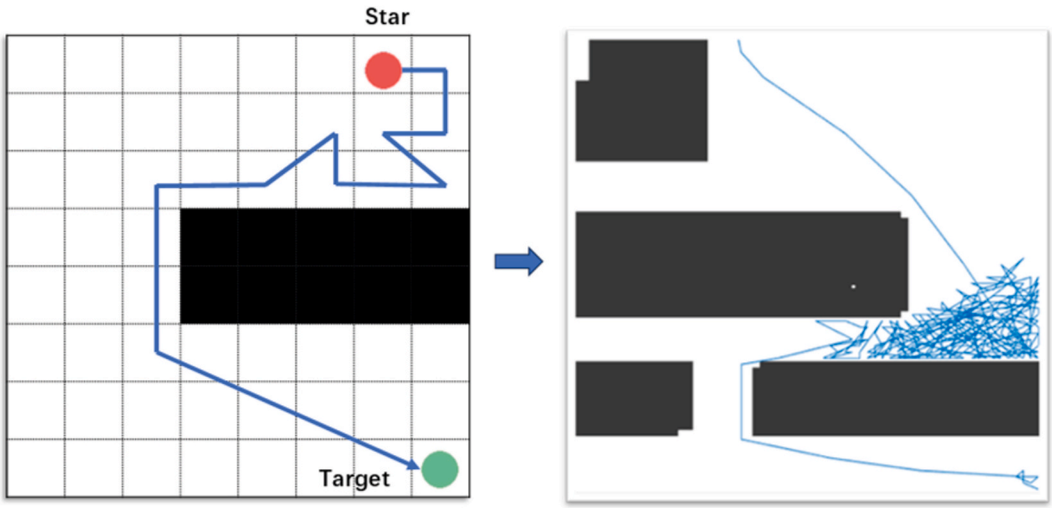



**Fig. 5.** The figure illustrates the detrimental effect when ants using the traditional heuristic formula enter the trap zone and become unable to escape.

pheromones across generations that they can ultimately plan the useable path (Wu and Du, 2025). However, in most existing ACO algorithms, pheromone updates and concentration occur only along the paths that ants travel. While this increases the differentiation within the ant colony, it limits the ants' search scope, thus restricting the algorithm's overall search capability and the optimization potential of other strategies.

### *4.1. New expansion strategy supporting the hierarchical decision-making elite ant optimization strategy*

The pheromones left by the pathfinding ants are not always in advantageous positions. Excessive pheromones can sometimes interfere with the ants' search ability. Ants may prematurely concentrate on a pheromone path, leading to early convergence or local optimum problems in the algorithm. A common optimization strategy is to introduce an elite ant system (Bai et al., 2024). Gao et al. ,2020 proposed the

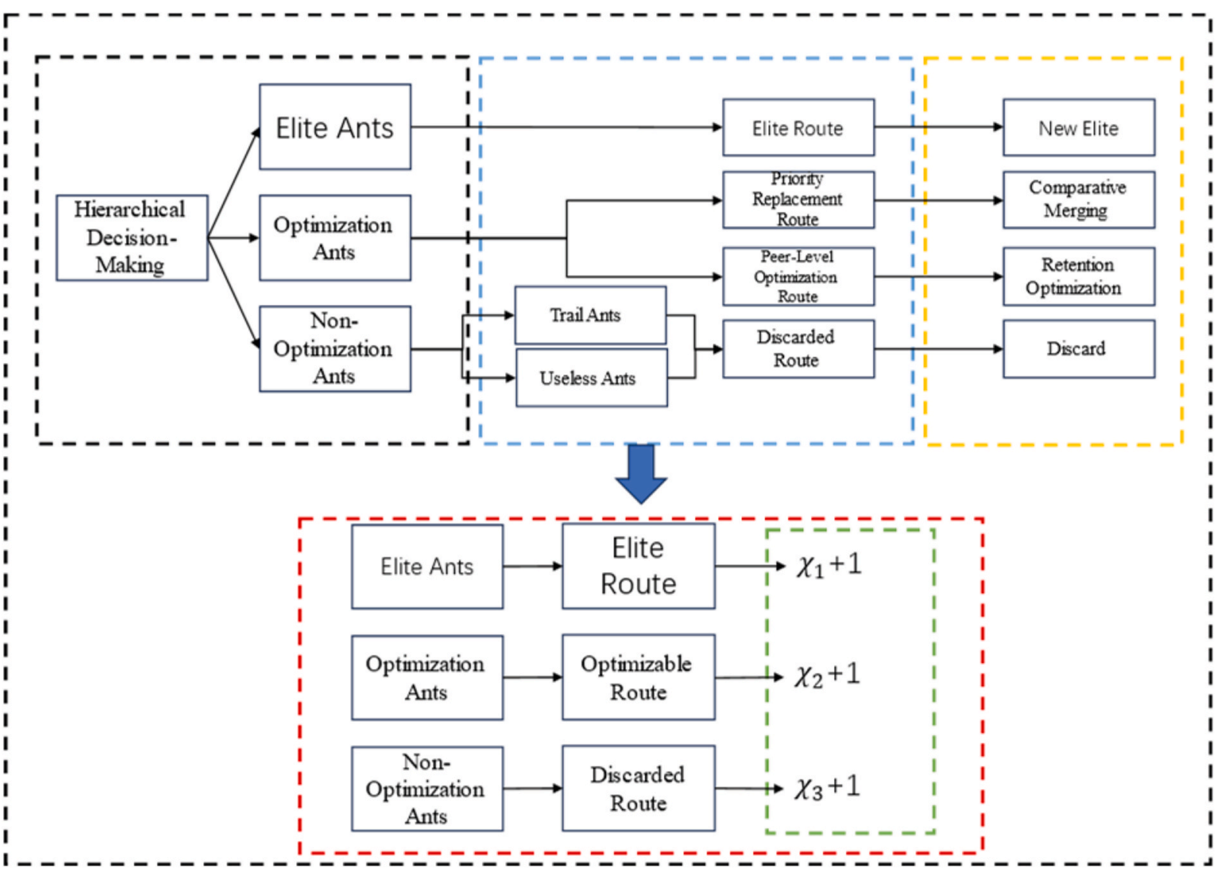



**Fig. 6.** The figure illustrates the specific scenario of the hierarchical decision-making elite ant strategy. The red box highlights the decisions made by the ants during the subsequent dynamic parameter adjustment. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

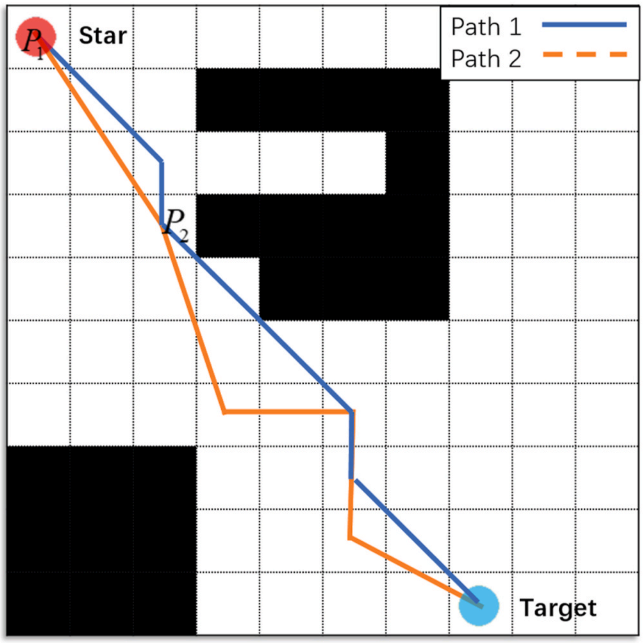



**Fig. 7.** Path1 represents the current locally optimal route left by the elite ants. Path2 represents a subsequently generated optimizable path. Although Path2 is not the shortest overall, it contains segments that can be further improved, such as the path between points $P_1$ and $P_2$.

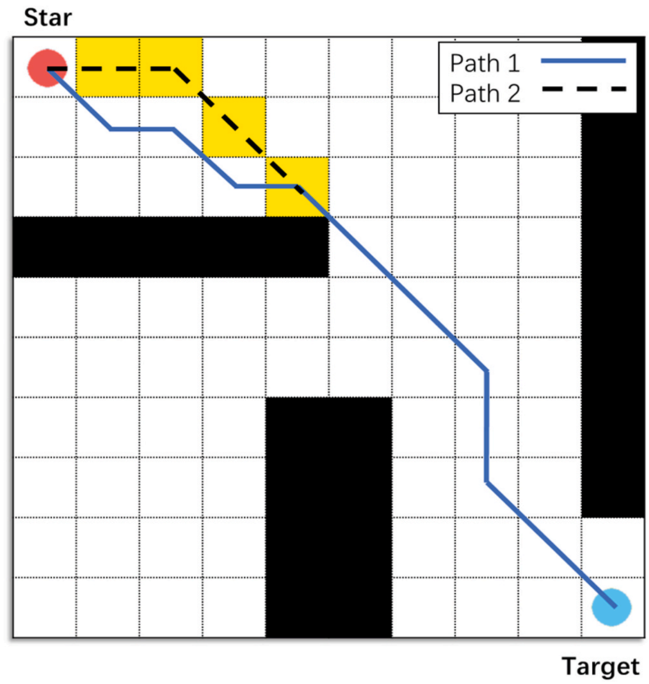



**Fig. 8.** Path 1 represents the generated elite path, while Path 2 illustrates an alternative optimizable route, referred to as the 'same-level optimization path.'"

EH-ACO algorithm, which integrates a direct path-merging strategy and a pheromone-merging mechanism. By expanding pheromone updates, it enhances the search area and optimizes local paths through merging operations. However, EH-ACO lacks a global identification mechanism for pheromone diffusion and a comprehensive discrimination mechanism for merging, resulting in limited effectiveness during experiments. In contrast, the PR-ACO algorithm proposed in this study introduces a novel and more robust strategy that provides a comprehensive and stable solution to path-planning challenges. The elite ant system optimizes the pheromone update mechanism by prioritizing elite ants, identifying the ants that generate the shortest paths during iterations as elite ants. The pheromone update is then applied only to the paths created by these elite ants, while the paths generated by non-elite ants are discarded. This method shows certain advantages as it makes the ant colony focus on the shortest path found, significantly accelerating the convergence speed. However, the drawbacks are also evident. The method completely discards the paths generated by non-elite ants, regardless of whether they may lead to better solutions in local areas during iterations. This weakens the optimization learning ability of the ACO algorithm, as ants may prematurely converge on suboptimal paths. Without optimization, this makes the ACO algorithm more prone to local optima. Additionally, by completely abandoning potentially optimized suboptimal paths, it leads to a significant waste of available computational resources, limiting the self-optimization capabilities of the ant colony. To address the aforementioned issues, this paper comprehensively improves the existing elite ant system and proposes a hierarchical decision-making elite ant strategy. Instead of treating all ants in the same manner as in traditional elite systems, this strategy distinguishes and classifies ants based on the characteristics of their routes in each iteration, applying differentiated handling accordingly. By incorporating swarm intelligence analysis and the concept of pheromone diffusion, this strategy establishes a novel, dynamic elite pheromone evaporation mechanism. In traditional elite ant systems, it is often observed that while some ants may generate suboptimal paths that are overall longer than the current elite path, certain segments of these routes may contain locally shorter and more efficient paths. Therefore, it is not advisable to discard such routes entirely. The hierarchical decision-making strategy proposed in this paper addresses this issue by evaluating and categorizing the routes generated by ants, allowing for more effective use of information from suboptimal paths.

This strategy classifies ants into three categories based on overall route quality: elite ants, optimizing ants, and non-optimizing ants, as illustrated in Fig 6. Elite ants generate the shortest path in the current iteration. Optimizing ants generate suboptimal paths that, although longer overall, may have shorter, better routes in certain areas. These paths can help improve the existing elite path. Non-optimizing ants fall into two categories: "useless ants" and "tracking ants." Useless ants generate paths that do not optimize the search and are significantly longer than other paths. Tracking ants strictly follow the elite ants' paths, reinforcing the pheromone on those routes. Both types are considered non-optimizing ants.

The routes generated by these ants can be classified into elite routes, optimizable routes, and discarded routes, as shown in Fig 6. Optimizable routes are further divided into "priority replacement routes" and "same-level optimization routes." Priority replacement routes are shorter than the elite route in certain areas. These local routes can replace the elite route and form a new elite path, as shown in Fig 7. It can be observed that although the Path2 has a longer overall length, it offers a better choice in local areas, and a merging strategy can be applied.

The same-level optimization strategy is applied when a route has a segment with the same length as the elite route. Since it's hard to judge its quality in the short term, a retention strategy is used. As shown in Fig. 8. Although the local path is the same length, it is smoother and more suitable for robotic kinematics and path smoothing strategies. Therefore, it should be retained for optimization. The obtained route undergoes the same pheromone enhancement strategy as the elite route within a set iteration time.

The improved scalability elite optimization strategy significantly aids in enhancing the iterative mechanism of the ant colony algorithm. However, the pheromone update strategy is a multifaceted mechanism that works in cooperation. The scalability elite optimization strategy primarily optimizes the retention of paths taken by the ants. While it has been effective, its optimization is more focused on local updates and lacks a broader global perspective. To better address the local optimality

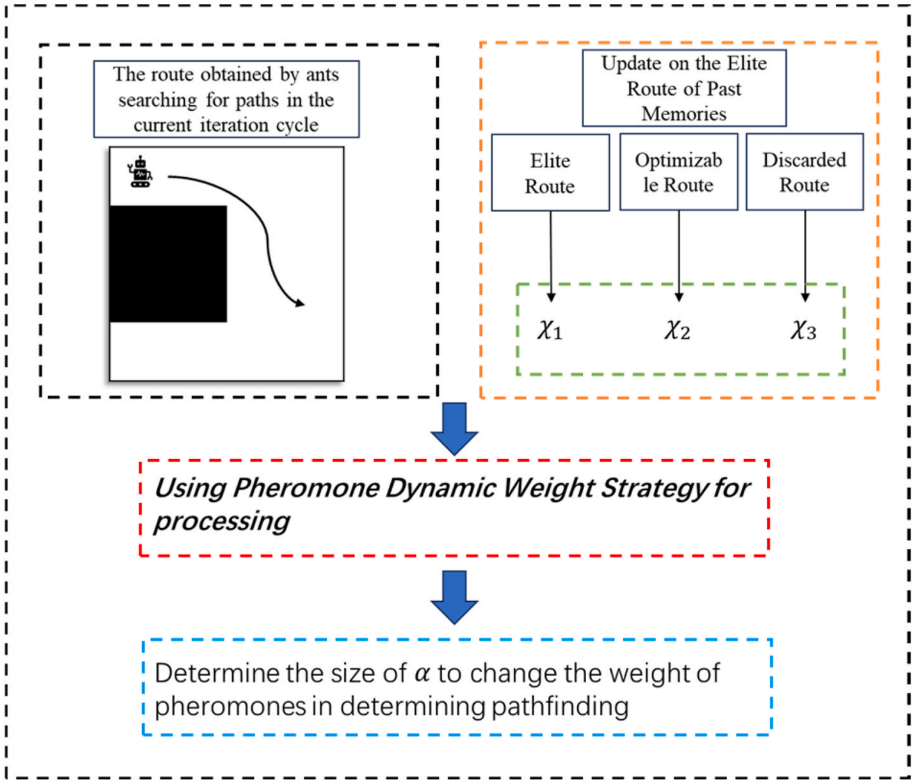



**Fig. 9.** Flowchart of using Pheromone Dynamic Weight Strategy.

and search efficiency issues, further optimization of another core aspect of the pheromone mechanism, the pheromone weight strategy, is necessary. The scalability elite optimization strategy mentioned earlier can also work well with the pheromone update strategy discussed below.

## 4.2. Pheromone dynamic weight strategy

Pheromones are adjusted through iteration, with their concentration increasing or decreasing along the ant paths, allowing pheromones to focus on the shortest paths. The pheromone weight determines the strength of the influence pheromones have on the ants. It is a key mechanism that enables ants to converge on the shortest path. Traditional pheromone weights follow a fixed rule, as shown in Equation (1), which outlines its application. It is evident that traditional pheromone weights are static, making pheromone responses slow to changes in iteration cycles and different path conditions. This is detrimental to the ant colony's ability to escape local optima. To address this issue, this study proposes a novel dynamic pheromone mechanism as a solution.

The scalability elite optimization strategy mentioned earlier significantly improves the utilization of iterative information and the flexibility of the ant colony at a local level. In some cases, the strategy may continuously optimize a suboptimal path, resulting in a better suboptimal path. However, due to the concentrated distribution of pheromones in ant colonies, there may be better global paths that have not yet been discovered. This paper proposes a novel dynamic pheromone weight updating strategy, as illustrated in Fig 9. This strategy utilizes information recorded by the newly constructed elite ant system described earlier to dynamically adjust the algorithm's weighting parameters. The formula for the dynamic pheromone weight update is defined as follows:

$$A = \chi_1 + \chi_2 \tag{12}$$

$$\alpha = \begin{cases} q_1 + q_2 \cdot \dfrac{1}{1 + e^{k(\sigma A - \chi_3)}} & t < e \\ q_1 + q_2 \cdot \dfrac{1}{1 + e^{-k(\chi_3 - \sigma A)}} & t \geq e \end{cases} \tag{13}$$

Here, $\chi_1$ refers to the number of elite ants that appear in the elite optimization strategy mentioned in this paper, while $\chi_2$ refers to the number of optimizing ants. The quantities of $\chi_1$ and $\chi_2$ can represent the number of effective optimizations performed by the algorithm. $\chi_3$ refers to the number of tracing ants in the non-optimizing ants group, and this quantity to some extent reflects the current number of times the ant colony converges to the old route. $t$ is the iteration count of the ant colony. The value of $e$ is generally equal to $t_{thresh}$ .These four parameters work together in a certain proportion, which can help determine whether the ants have prematurely converged due to local optima or have already found the optimal path and are converging. $q_1$ is a limiting parameter to prevent the value of $\alpha$ from being too small or too small. $q_2$ is a weight parameter.

$\sigma$ is a proportion used to control whether the output should decrease or increase. $k$ controls the rate of change; a larger $k$ makes the change steeper, while a smaller $k$ makes it smoother. In the early stages of iteration, when $t < e$, if $A$ is too small and $\chi_3$ is too large, the ant colony is highly likely to get trapped in a local optimum. In this case, as $\chi_3$ increases, $\alpha$ should decrease within an appropriate range to give the ant colony better global vision and encourage it to search for better paths.

When $t > e$ in later iterations, as $\chi_3$ increases, $\alpha$ should be increased within an appropriate range to make the ant colony focus more on the optimal path it has already found. This helps prevent the colony from blindly searching and encountering the "reversal phenomenon," where the colony diverges by searching longer paths unnecessarily, leading to convergence fluctuations.

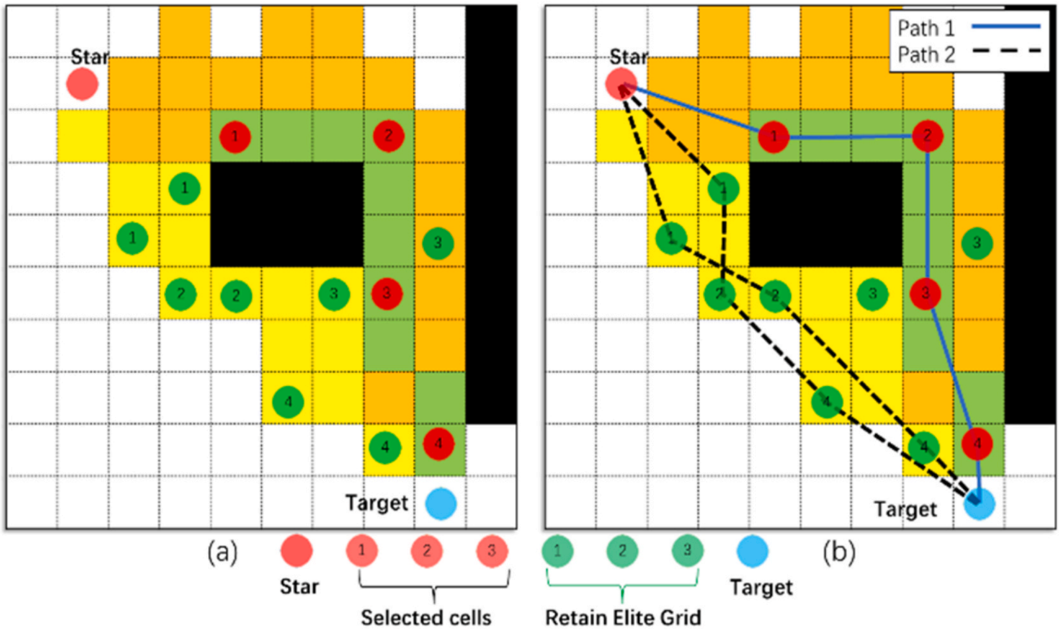



**Fig. 10.** (a) Illustrates the distribution of pheromones. (b) Illustrates the distribution of pheromones and the corresponding generated path, as well as potential favorable paths that may emerge in future iterations.

### 4.3. Pheromone topological diffusion strategy

As mentioned earlier, in most existing ACO algorithms, pheromone updates and concentrations only occur along the paths traveled by the ants. This approach is not well-suited for the wide-area search strategy proposed in this paper. The wide-area search strategy generates ant colony paths in an unconventional manner, starting with dispersed points within an expanded search field and then connecting them into paths. Consequently, the initial pheromone distribution is not continuous but concentrated at specific points, making it harder for other points along the optimal path to be selected during iterations. This limitation affects the convergence capability of the ants.

Inspired by the RRT algorithm's topological diffusion strategy, this paper proposes a new pheromone diffusion mechanism for retaining elite points. This mechanism enhances the global search performance of the pheromone, improves the handling of local optima, and strengthens the convergence capability of the ant colony algorithm. The specific formula is as follows:

$$d_w = d(g, g_i) + d\left(g, g_j\right) \tag{14}$$

$$\nabla\gamma = \begin{cases} 1, & d\left(g_i, g_j\right) < d_w \le 1.2d\left(g_i, g_j\right) \\ 0.25, & d\left(g_i, g_j\right) < d_w \le \sqrt{2}d\left(g_i, g_j\right) \\ 0, & other \\ 0.25*\dfrac{u}{d\left(g_i, g_j\right) + t} & d_w \le 1.2d\left(g_i, g_j\right), \left(g_j \notin PL_t\right) \end{cases} \tag{15}$$

Here, $g_i$ represents the grid where the ant is currently located, $g$ refers to the grid being evaluated, and $g_j$ denotes the next grid that has been determined or an elite retention grid that is available but not selected. $d(g, g_i)$, $d\left(g, g_j\right)$, and $d\left(g_i, g_j\right)$ indicate the distances between two grids. The ant sequentially diffuses pheromones along the path $PL_t\left(g_1, g_2, g_3 \dots g_i, g_j \dots g_{n-1}, g_n\right)$ it has determined so far.

If $j$ is not part of the currently confirmed path $\left(g_j \notin PL_t\right)$, it can be identified as a retained elite point grid $g_j$, which will then enter the final judgment condition of Equation (15). $u$ represents the number of retained elite points, and these grids will diffuse pheromones with slightly weaker intensity.

Based on this integration, the new formula for pheromone intensity is as follows:

$$\Delta\tau_i(t)^k = \frac{Q_1}{L_{best}(t)}\Delta\gamma \tag{16}$$

Pheromone diffusion is implemented by spreading between two selected grid points in a topological extension manner, as shown in Fig 10. This effectively addresses the issue of uneven pheromone distribution. Green grids represent the topological grids of selected points, where pheromone concentration increases significantly. Orange grids represent the pheromone diffusion grids of selected points with moderate pheromone increase. Yellow grids denote the pheromone diffusion grids of retained elite points with the smallest increase in pheromone concentration.

This controlled diffusion increases the likelihood of the ant colony escaping local optima. In cases where the optimal path is not found in the current iteration, the algorithm gains greater potential for optimization in subsequent iterations.

Path1 represents the confirmed path in the current iteration, which is evidently not optimal. Path2 represents a potentially better path that may emerge in the next iteration. This approach not only improves the coverage of pheromones at locations where optimal paths are likely to appear but also enhances the colony's ability to search for superior paths. It addresses the local optimality issue of the ACO algorithm while balancing convergence performance and improving global planning capabilities.

In Fig 10 (b), Path1 represents the current generated path, while Path2 indicates a potential path for the next iteration based on the pheromone distribution. In Fig 10,Grid cells marked with red circles represent currently selected points. Grid cells with green circles represent retained elite points, which, although not selected, exhibit favorable indexing and will be preserved for one iteration. The numbers indicate the step sequence of the ant's movement; for example, "1″ represents the three grid cells visited during the ant's first step.

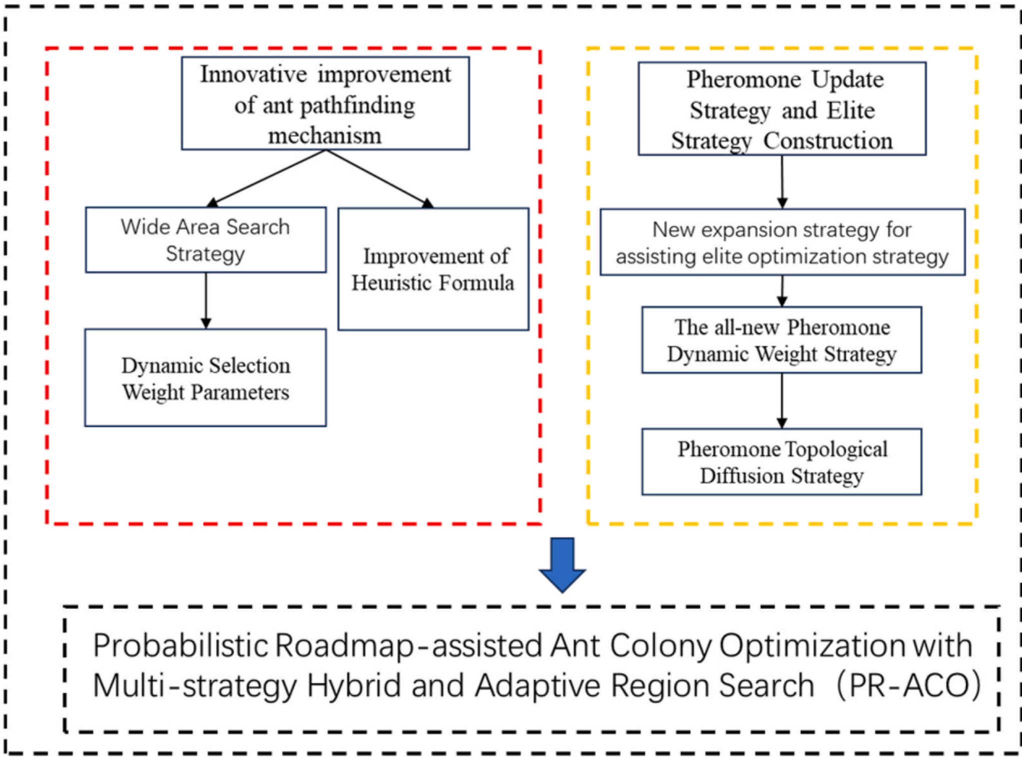



**Fig. 11.** Presented the workflow for improving the PR-ACO algorithm in this study.

## 5. Algorithm usage steps

In Chapters 4 and 5, this paper designs the PR-ACO algorithm to improve the traditional ACO algorithm model. Detailed planning is provided for the framework processes of strategies such as the Wide Area Search Strategy, Improvement of Heuristic Formula, and Pheromone Update Strategy and Elite Strategy Construction. The flowchart shown in Fig 11 illustrates the detailed working process of the proposed PR-ACO algorithm. In this study, both the pathfinding mechanism and the learning capability of the traditional ACO algorithm are innovatively optimized. The improved PR-ACO algorithm demonstrates significant improvements in path length, planning time, and route smoothness. Additionally, the specific steps of the improved algorithm are presented in the form of pseudocode to clearly illustrate the implementation process. Ultimately, based on the task sequences generated by ACO, the improved PR-ACO algorithm produces superior collision-free paths. The final improved algorithm flow is shown in **Algorithm I**.

## 6. Experimental verification

This chapter tests the proposed PR-ACO algorithm using a MATLAB program and a physical experimental platform built on the ROS system. The simulation experiments were conducted on a computer equipped with an Intel Core i5-12th Gen 2.50 GHz CPU and 16 GB RAM, with the simulation implemented in MATLAB R2024a.The physical experimental platform is based on the TurtleBot2 robot platform, equipped with an industrial computer, a LiDAR sensor, and an Xbox depth camera.

### *6.1. Introduction to simulation experiment*

In the simulation experiments, to ensure the validity of the results, in addition to comparing with the traditional ACO algorithm, this paper also reproduces the EH-ACO algorithm from Gao et al. ,2020 and the D-ACO algorithm proposed by Dai et al., (2018). Since the EH-ACO algorithm is based on the traditional search mechanism, it introduces a locally focused heuristic function and a new local pheromone diffusion and merging mechanism that consider the distance between the current point and the goal. Comparing with EH-ACO shows how PR-ACO's wide-area search, new heuristic, and pheromone update mechanism improve adaptability across maps. D-ACO targets path smoothness, but its limited search scope may fail on large maps. PR-ACO vs. D-ACO highlights PR-ACO's effectiveness in smoother paths and updated pheromone strategies. These reproduced algorithms are compared with the proposed PR-ACO algorithm across different map environments. To verify the algorithm's adaptability, the maps are divided into three categories: large, medium, and small. Targeted experiments are conducted on these three types of maps. The four map environments are referred to as Environment 1, Environment 2, Environment 3, and Environment 4.

**Table 1**
Important parameters shared by various algorithms.

| | K | M | $\alpha$ | $\beta$ | $Q_1$ | $l$ | $\varepsilon$ |
|---|---|---|---|---|---|---|---|
| PR-ACO | 50 | 20 | 1 | 7 | 1.3 | 1 | 4 |
| ACO | 50 | 50 | 1 | 7 | 1.3 | 1 | – |
| EHACO | 50 | 20 | 1 | 7 | 1.3 | 1 | – |
| D-ACO | 50 | 20 | 1 | 7 | 1.3 | 1 | – |

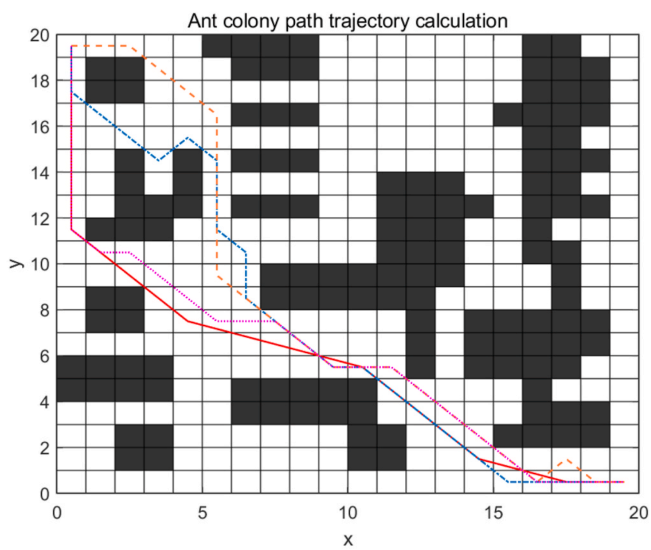
Environment 1 demonstrates the performance of various algorithms on complex small maps, primarily testing the adaptability of the PR-ACO algorithm on small maps. Environment 2 represent medium to large maps, testing the PR-ACO algorithm's performance in handling complex, large-scale environments. The conditions of Environment 3 and Environment 4 are more complex, which will be discussed separately in Section 6.4.

To ensure fairness in algorithm testing, this paper ensures that all algorithms share the same parameters, guaranteeing the smooth operation of each algorithm. The key parameters include the total number of iterations K, the number of ants M per iteration, the importance of the heuristic formula, and the significance of pheromone representation, along with PR-ACO-specific parameters such as $\varepsilon$, as shown in Table 1.

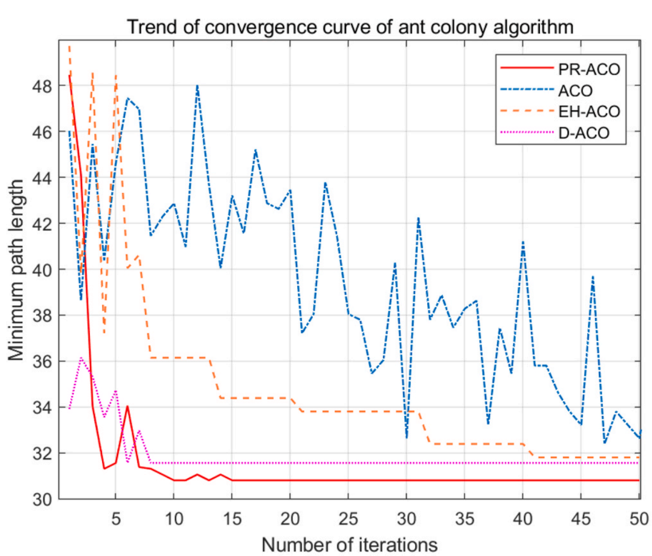
**Table 2**
Important parameters in PR-ACO algorithm.

| | $q_1$ | $q_2$ | $r_{\min}$ | $t_{thresh}$ | $C$ | $C_1$ | $p$ | $a_{\min}$ | $a_{\max}$ |
|---|---|---|---|---|---|---|---|---|---|
| PR-ACO | 1 | 0.3 | 2 | 10 | 15 | 0.9 | 0.2 | 0.5 | 5 |

Experiments show that the optimal range for $\varepsilon$ is typically between number (3–8). Smaller values of $\varepsilon$ tend to yield better performance in most cases. In this paper, we adopt a relatively aggressive value within that range to highlight the improvements brought by the proposed method. Table 2 lists the key parameters specific to the proposed PR-ACO algorithm. The value of $r_{\min}$ is constrained by the grid resolution. In our experiments, the grid unit length is set to 1, making the optimal range for $r_{\min}$ between number (1–2.5); here, we select $r_{\min} = 2$. As for $t_{thresh}$, experimental results show that values between number (6–12) offer a good trade-off between performance and stability. To better assess the algorithm's enhancements, we set $t_{thresh} = 10$ in simulation tests. The parameter $a_{\min}$ is set to 0.5. Parameter $a_{\max}$ also delivers satisfactory results and can be adjusted based on specific needs in different scenarios. Other balancing constants such as $C$ and $C_1$ are determined according to predefined values like initial pheromone concentration and evaporation rate within the experimental codebase. For instance, $C$ is mainly used to dilute the weight of the heuristic function relative to the pheromone concentration. In our experimental environment, $C$ values between number (10–20) demonstrate relatively stable performance, while values outside this range tend to be less favorable. Since this study uses a grid-based map analysis, the grid parameter $l$ is set to 1 but can be adjusted as needed; larger $l$ reduces map resolution.




(a)




(b)

**Fig. 12.** (a) Shows the path results of each algorithm after iteration. (b) Provides detailed iteration information for each algorithm. The legend in the top right corner of figure (b) corresponds to the lines of each algorithm, and (a) and (b) share this legend.

**Algorithm 1.** The process of PR-ACO.

Algorithm I

Grid method for constructing spatial models
Initialize parameters such as $M$, $K$, $\alpha$, $\beta$, $r_{\min}$, $k$, $p$, $\varepsilon$, $k_1$, $k_2$, $C$, $C_1$, $q_2$, $t_{thresh}$, $Q_1$ and others, along with the path matrix and pheromone matrix.
  **For** k = 1 to K do;
    **For** m = 1 to M do;
    Initialize ants;
    Initialize search radius;

$$r = \max\left(r_{\min}, \frac{1}{\varepsilon}\cdot\sqrt{a^2+b^2}\cdot(1-k\cdot p)\right)$$

    **While** Ant has not reached the destination;
    Randomly select a reachable node;

$$P_j = \frac{\lambda_j}{\sum_{s\in allow_k}\lambda_j}\lambda_j = \left[\frac{1}{1+e^{-(t-t_{\text{thresh}})}}\cdot\left(\frac{n_j^{0.5}}{a_{\max}}\right) + \left(1 - \frac{1}{1+e^{-(t-t_{\text{thresh}})}}\right)\cdot\left(\frac{a_{\min}}{n_j^{0.5}}\right)\right]\cdot k_1 + k_2\cdot\tau_j$$

$$\eta_{ij}(t) = \frac{1}{\frac{d\left(g_i,g_j\right)}{r} + \left(1+\frac{C_1}{d\left(g_i,g_j\right)}\right)*d\left(g_j,g_{\text{target}}\right) + C}$$

$$P_{ij}^k(t) = \begin{cases} \frac{\tau_j{}^{\alpha}(t)\cdot\eta_{ij}^{\beta}(t)}{\sum_{s\in allow_k}\tau_j{}^{\alpha}(t)\cdot\eta_{ij}^{\beta}(t)}, & s\in allow_k \\ 0, & s\notin allow_k \end{cases}$$

  **End** While
  **End** for
    **If** the ant reaches the destination;
      **If** it is an elite path:;
        Update the elite path;
      **Else**
        Determine whether the route is feasible;
        **If** there are optimizable points;
          Execute merging strategy or retention strategy;
        **End** if
    Update the pheromone of the obtained elite path;

$$A = \chi_1 + \chi_2$$

$$\alpha = \begin{cases} q_1 + q_2\cdot\frac{1}{1+e^{k(\sigma A-\chi_3)}} & t < e \\ q_1 + q_2\cdot\frac{1}{1+e^{-k(\chi_3-\sigma A)}} & t \geq e \end{cases}$$

$$\nabla\gamma = \begin{cases} 1, & d\left(g_i,g_j\right) < d_w \leq 1.2d\left(g_i,g_j\right) \\ 0.25, & d\left(g_i,g_j\right) < d_w \leq \sqrt{2}d\left(g_i,g_j\right) \\ 0, & other \\ 0.25*\frac{u}{d\left(g_i,g_j\right)+t} & d_w \leq 1.2d\left(g_i,g_j\right), \left(g_j \notin PL_t\right) \end{cases}$$

$$\Delta\tau_i(t)^k = \frac{Q_1}{L_{best}(t)}\Delta\gamma$$

$$\tau_i(t+1) = (1-\rho)\tau_i(t) + \Delta\tau_i(t)$$

  **End** if
**End** for
Out put optimal path;

### 6.2. Environment 1 Experimental results

In Environment 1, three algorithms— ACO, EH-ACO, and D-ACO—were used to compare the performance of the PR-ACO algorithm. The map size of Environment 1 is 20*20. The path results of each algorithm are shown in Fig 12(a). It can be observed that the PR-ACO

**Table 3**
Statistics of the iterative results of each algorithm. Average time refers to the average time required for each algorithm to run ten times.

| | Iterations | Path length | Turning times | Average time | Variance of time |
|---|---|---|---|---|---|
| PR-ACO | 15 | 30.8005 | 5 | 1.209 | 0.1144 |
| ACO | 50 | 32.6274 | 10 | 1.613 | 0.0422 |
| EH-ACO | 41 | 31.799 | 8 | 1.119 | 0.1287 |
| D-ACO | 8 | 31.556 | 8 | 1.164 | 0.1285 |

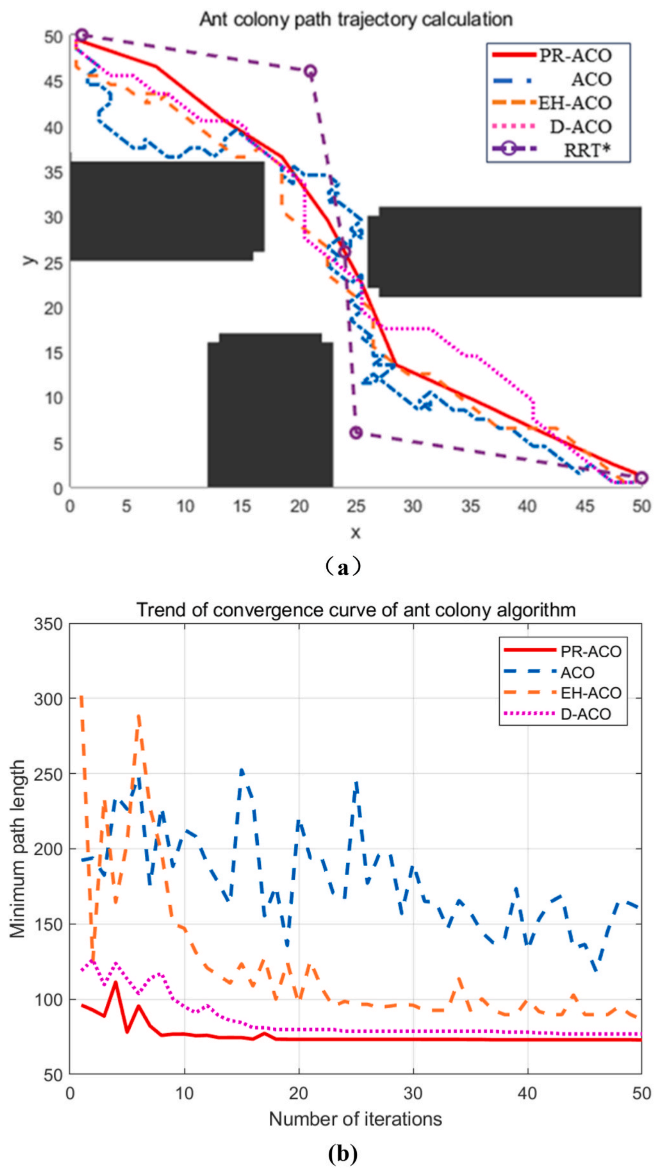



**Fig. 13.** (a) Shows the path results of each algorithm after iteration. (b) Provides detailed iteration information for each algorithm. The legend in the top right corner of figure (b) corresponds to the lines of each algorithm, and (a) and (b) share this legend. The RRT * algorithm does not have an iterative curve, so it is not shown in (b).

algorithm produces a path with fewer turns and better overall performance compared to the other algorithms. From Fig 12(b) and Table 3, it is evident that the PR-ACO algorithm yields a shorter path, as its broader search field results in a more global path with fewer turns, making it more applicable in practical scenarios. This demonstrates the effectiveness of the dynamic parameters in balancing global search. In contrast, the D-ACO algorithm converged at the 8th generation, but its path was not as optimal in terms of path length and number of turns. The premature convergence led the algorithm into a local optimum, which becomes more significant in larger maps.

**Table 4**
Statistics of the iterative results of each algorithm. (The "Average Time" values in the table are from MATLAB simulations and mainly reflect runtime percentage differences. In real-world experiments, the actual times are expected to be much smaller.)

| | Iterations | Path length | Turning times | Average time | Variance of time |
|---|---|---|---|---|---|
| PR-ACO | 18 | 73.4216 | 5 | 33.099 | 2.9151 |
| ACO | 50 | 159.723 | – | 40.571 | 7.8667 |
| EH-ACO | 50 | 86.5685 | 16 | 36.672 | 3.1828 |
| D-ACO | 43 | 76.9117 | 20 | 37.492 | 3.0530 |
| RRT* | – | 81.526 | 3 | 59.461 | 4.0474 |

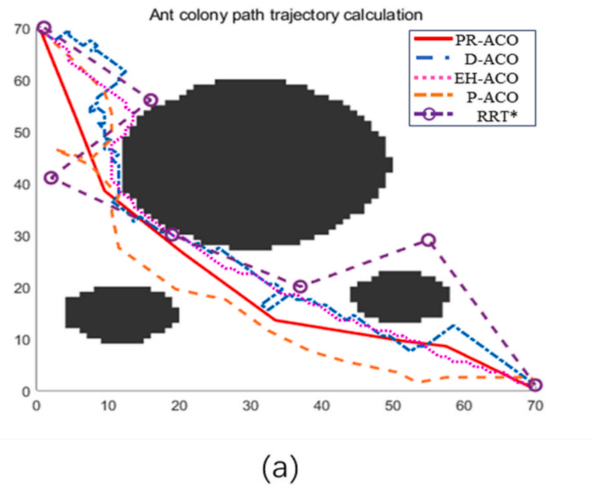
As shown in Table 3, the runtime comparison among algorithms indicates that on small-scale maps, PR-ACO's execution time is slightly longer—7.4 % longer than EH-ACO and 3.3 % longer than D-ACO. However, PR-ACO achieves the shortest path length and the fewest turning points, demonstrating significantly better overall performance. For instance, compared with D-ACO, PR-ACO reduces path length and the number of turns by 2.4 % and 37.5 %, respectively. Compared with EH-ACO, it shortens the path length by 3.14 %. Furthermore, under the same conditions, PR-ACO runs 25 % faster than the original ACO algorithm. These results indicate that although PR-ACO introduces slightly higher computational complexity than some variants, the overall performance gains are clearly worthwhile. By comparing the variance in runtime, it can be seen that although the differences are small, PR-ACO still demonstrates consistently superior performance. Traditional ACO, while showing the smallest variance, generally requires longer runtimes, and its runtime fluctuates significantly on larger maps. Additionally, in subsequent experiments on larger maps, PR-ACO also achieved reduced runtime, further confirming its scalability and efficiency.

### 6.3. Environment 2 Experimental results

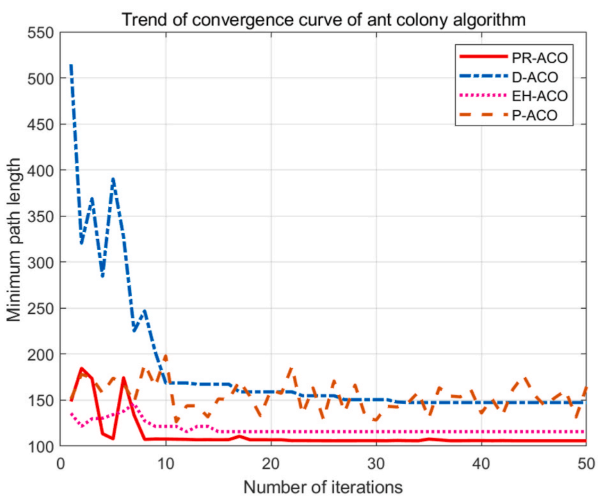
In Environment 2, the map size is 50*50 grids, resulting in a total of 2500 grids, which is six times larger than Environment 1. In this case, some issues become more prominent. The four algorithms (PR-ACO, ACO, EH-ACO, and D-ACO) are used again to compare the performance of PR-ACO. In addition, to demonstrate the applicability of PR-ACO, this chapter also includes a comparison with the RRT* algorithm to evaluate their respective path generation capabilities. Experiments show that the step size for the RRT* algorithm should be set between 15 and 20 and adjusted according to map size. In Environment 2, RRT* was configured with a step size of 15, a maximum of 10,000 iterations, and a goal acceptance radius of 10.

From Fig 13(a) and (b), it can be seen that the advantage of PR-ACO becomes more apparent in the medium-sized map. The improved algorithm produces a much smoother path compared to the other algorithms. From Table 4, it is evident that PR-ACO not only significantly improves path length but also shows a considerable reduction in iteration convergence time and turning points. These results demonstrate the effectiveness of the algorithm's dynamic wide-area search strategy.

From Fig 13(b), it can be observed that the PR-ACO algorithm converges at the 18th generation, but it reaches a relatively suboptimal path by the 8th generation. Between generations 8 and 18, the algorithm undergoes several fine-tuning adjustments, ultimately arriving at the optimal route. This demonstrates the advantage of the algorithm's global optimization capability. On the other hand, ACO, EH-ACO, and D-ACO perform less effectively. EH-ACO and ACO algorithms fail to




(a)




(b)

**Fig. 14.** (a) Shows the path results of each algorithm after iteration. (b) Provides detailed iteration information for each algorithm. The legend in the top right corner of figure (b) corresponds to the lines of each algorithm, and (a) and (b) share this legend.

successfully converge. D-ACO reaches local convergence at the 18th generation but requires an additional optimization in the 43rd generation, still resulting in a poor overall outcome and getting stuck in a local optimum.

By comparing the data in Tables 4 and it is evident that PR-ACO demonstrates superior adaptability and performance on larger-scale maps. In Environment 2, the average runtime of PR-ACO is approximately 9.74 % shorter than EH-ACO and 12.1 % shorter than D-ACO, while the generated paths are also significantly improved. Comparing the variance shows that traditional ACO exhibits large fluctuations in runtime due to frequent failures to find paths on time, resulting in high variance. In contrast, PR-ACO demonstrates minimal runtime variance, highlighting its stability. This is mainly because, in large-scale environments, algorithms such as EH-ACO and D-ACO suffer from ants having limited search vision, which can lead to disoriented or inefficient pathfinding. As a result, the time for individual ants to find a path increases, and the overall convergence of the algorithm slows down—ironically increasing the algorithm's complexity. When compared with the RRT* algorithm, although RRT* can produce relatively effective paths, this performance is achieved only by substantially increasing the tree growth step size and the number of iterations, leading to excessive runtime. Furthermore, the resulting paths are still not optimal.

**Table 5**
Statistics of the iterative results of each algorithm.(Average time is used for percentage comparison).

| | Iterations | Path length | Turning times | Average time | Variance of time |
|---|---|---|---|---|---|
| PR-ACO | 37 | 105.894 | 3 | 117.394 | 4.1321 |
| D-ACO | 32 | 147.966 | 51 | 177.335 | 5.2191 |
| EH-ACO | 15 | 115.782 | 45 | 167.117 | 4.5027 |
| P-ACO | 50 | 121.581 | 11 | 121.394 | 4.7610 |
| RRT* | – | 121.404 | 4 | 181.912 | 4.2825 |

Overall, PR-ACO demonstrates a strong advantage in this map environment.

### 6.4. Environment 3 Experimental results

In Environment 3, a larger grid map of 70x70 is used, expanding several times over compared to Environment 1 and Environment 2. This map is designed to test the performance of the PR-ACO algorithm on large-scale maps, highlighting convergence and local optimum issues. A new challenge arises here, as the traditional ACO algorithm fails to plan a path. During pathfinding, multiple instances occur where ants cannot find food, rendering it unsuitable for experimental comparison. Therefore, a new comparison group, P-ACO, was introduced. P-ACO inherits some heuristic innovations from D-ACO while incorporating the same wide-area search radius as PR-ACO. Comparing P-ACO with EH-ACO highlights the advantages of wide-area search over traditional eight-direction search. Meanwhile, comparing P-ACO with PR-ACO demonstrates the adaptability and effectiveness of the new heuristic function and the updated pheromone merging mechanism proposed in this study. This ensures a fairer comparison, as both algorithms use the same search radius, isolating the influence of other modifications and allowing for a more accurate assessment of the wide-area search strategy effectiveness. To ensure a comprehensive and fair comparison, the RRT* algorithm—commonly used in robot path planning—was also selected as a baseline for evaluation alongside PR-ACO in this study. In the Environment 3 experiment, the RRT* algorithm used a step size of 17, a maximum of 11,000 iterations, and a goal tolerance of 10.

P-ACO incorporates the dynamic radius from D-ACO, aligning with the wide-area search theory described in Section 4.1. This comparison helps highlight the superiority of the wide-area search theory over traditional 8-direction search methods. As shown in Fig 14(a), the path generated by the P-ACO algorithm is significantly optimized compared to D-ACO, with fewer sharp turns and smoother paths. While the overall path still requires optimization, it clearly demonstrates global performance characteristics.

However, Fig 14(b) shows that the P-ACO algorithm struggles with convergence. This is because P-ACO lacks the subsequent dynamic selection of weight parameters and pheromone updates that PR-ACO uses, resulting in poor convergence and affecting its pathfinding ability.

From Fig 14(b) and Table 5, it is evident that PR-ACO performs excellently on the large map. While it appears to fully converge by the 37th generation, Fig 14(b) shows that PR-ACO actually finds a relatively optimal path (with a length of 107.337) by the 8th generation. In the subsequent iterations, the algorithm does not lose its optimization ability; it continues to refine the path while maintaining convergence, ultimately arriving at the optimal path. This demonstrates PR-ACO's exceptional global performance and self-optimization capability.

On the other hand, the EH-ACO algorithm, though it converges by the 15th generation, generates a suboptimal path due to premature convergence, as seen in Fig 14(a). As shown in Table 5, the paths generated by the PR-ACO algorithm are significantly superior in both length and smoothness compared to those produced by other algorithms. Specifically, when compared to the EH-ACO algorithm, which

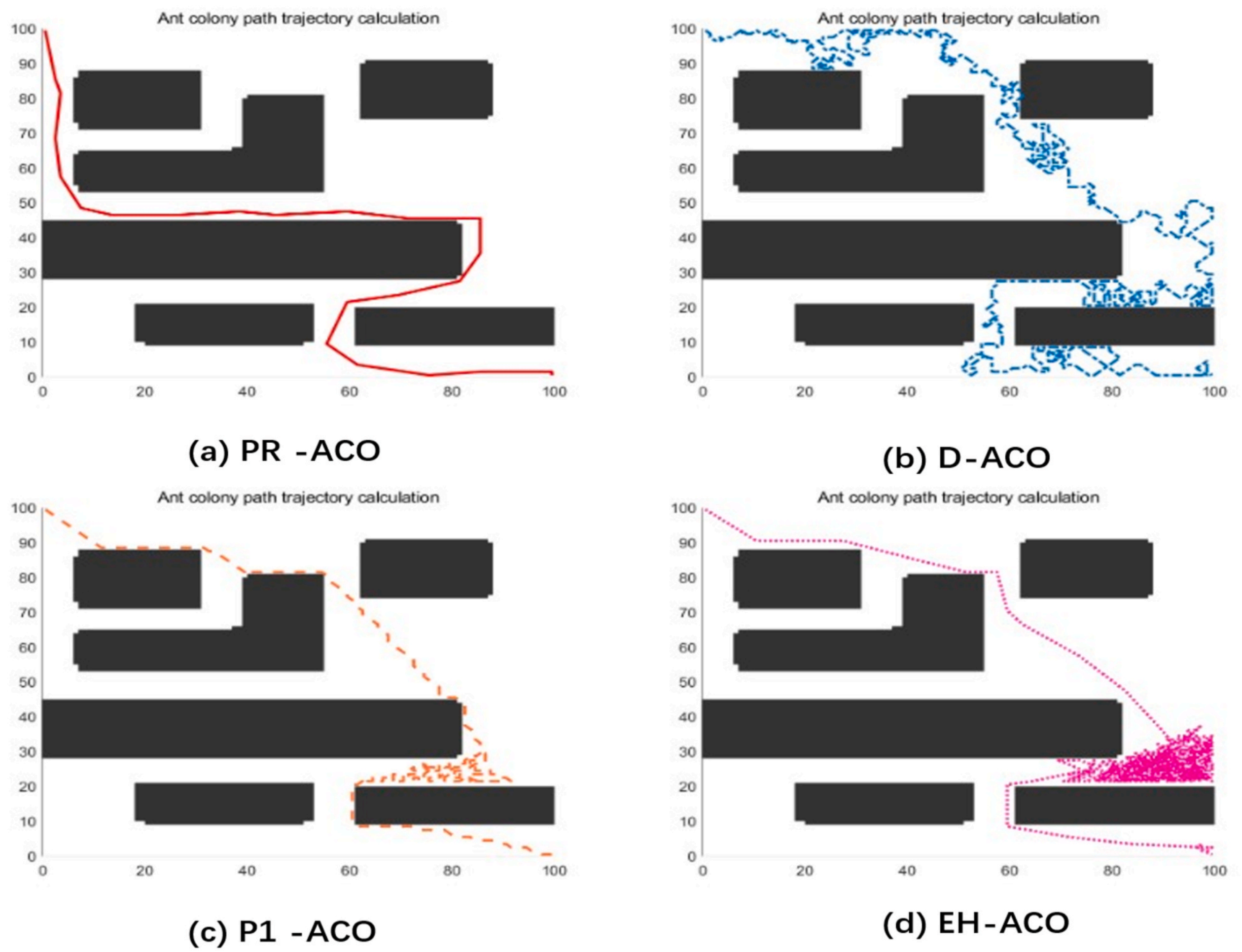



**Fig. 15.** (a) Represents the path trajectory of the PR-ACO algorithm. (b) Represents the path trajectory of the D-ACO algorithm.(c) represents the path trajectory of the P1-ACO algorithm. (d) Represents the path trajectory of the EH-ACO algorithm.

yields paths with numerous sharp turns, the number of turning points in PR-ACO's paths is reduced by 93.3 %, and the path length is shortened by 8.5 %. The shortcomings of the D-ACO algorithm are even more pronounced, as it produces paths with both significantly greater length and more frequent turning points.

The algorithmic complexity can be assessed by analyzing the average runtime of each algorithm, as shown in Table 5. On this map, PR-ACO fully demonstrates its strength in adaptability, with its runtime reduced by 33.9 %, 29.9 %, 3.3 %, and 35.3 % compared to D-ACO, EH-ACO, P-ACO, and RRT* respectively. For RRT* to generate better-quality paths in this environment, it would require a further increase in its tree extension step size and iteration count, which significantly prolongs the computation time. Moreover, RRT* already performs worse than PR-ACO in terms of path length and overall path quality on this map. Although P-ACO benefits from a broad-area search strategy that effectively reduces runtime, it lacks additional optimization mechanisms, resulting in less smooth paths that still leave room for improvement. Based on the variance of average runtime, PR-ACO shows the smallest variance, demonstrating stable performance on larger maps. In contrast, P-ACO, which lacks the improved heuristic function and partial pheromone update mechanisms of PR-ACO, occasionally performs well in pathfinding but exhibits much lower stability. In summary, PR-ACO exhibits clear advantages, generating shorter paths with fewer turns, thereby offering both higher efficiency and better path quality.

### *6.5. Environment 4 Experimental results*

The map of Environment 4 utilizes a more complex large-scale environment within a 100 × 100 area, featuring multiple obstacles, including trap regions, which pose a significant challenge to the overall performance of the algorithm. Traditional ACO algorithms can no longer effectively plan routes in this scenario. To specifically evaluate and compare the performance of the PR-ACO algorithm, a new comparison algorithm, P1-ACO, was introduced. In this section, four algorithms—PR-ACO, P1-ACO, EH-ACO, and D-ACO—are compared. The P1-ACO algorithm is a test version based on PR-ACO, lacking the improvements in the heuristic formula described in Section 4.3. But it uses the heuristic formula of EH-ACO algorithm. Its primary purpose is to test whether the improvements in the heuristic formula can handle complex environments and solve the "trap region" problem. Therefore, the parameters of the P1-ACO algorithm are identical to those of PR-ACO and follow the parameters listed in Table 1.

As shown in Fig 15 the PR-ACO algorithm performs exceptionally well in highly complex large-scale maps. In contrast, the EH-ACO and D-ACO algorithms have largely lost their ability to search for optimal paths in broader environments. As illustrated in Fig 15(d), while the EH-ACO algorithm can still plan basic routes in the Environment 3 map, it is no longer able to identify the optimal route.

As depicted in Fig 15(b), (c), and (d), all algorithms except PR-ACO are unable to escape from trap regions to varying degrees, ultimately converging on paths that are not useable. Additionally, Fig 16 shows that in maps with even greater complexity and scale, the EH-ACO and D-

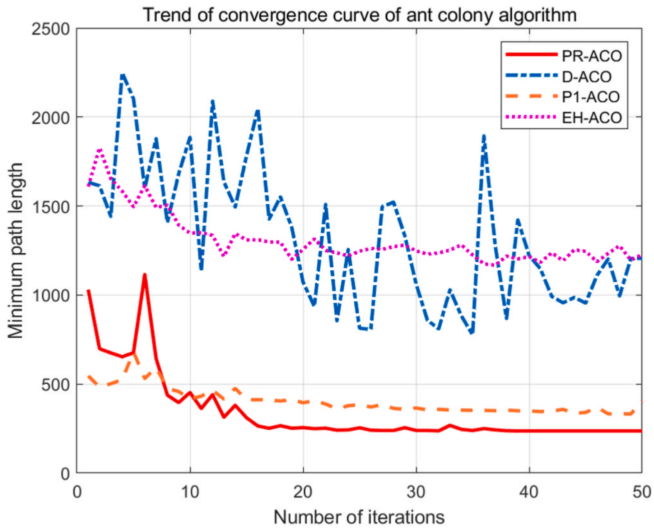



**Fig. 16.** The detailed information of the iteration process for each algorithm is illustrated. The legend in the upper-right corner corresponds to the lines representing each algorithm, and this legend is shared between.

**Table 6**
Statistics of the iterative results of each algorithm.

| | Iterations | Path length | Average time |
|---|---|---|---|
| PR-ACO | 38 | 237.717 | 336.532 |
| D-ACO | 50 | 776.164 | 493.702 |
| P1-ACO | 47 | 332.622 | 397.950 |
| EH-ACO | 50 | 1187.63 | 517.091 |

ACO algorithms fail to converge successfully. Although the P1-ACO algorithm has made some optimizations, it is limited by the constraints of the heuristic formula and is ultimately unable to completely escape local optima. The P1-ACO algorithm achieves initial convergence at the 15th generation and complete convergence by the 47th generation. Throughout the process, it maintains its optimization capability, attempting incremental improvements in the path with each generation. Similarly, the PR-ACO algorithm also performs optimization attempts, achieving initial convergence at the 17th generation with a path length of 251.596 and ultimately finding the shortest path in the 38th generation, as shown in Table 6 and Fig 16.

As shown in Fig 15(a), (c), and Fig 16, the comparison between PR-ACO and P1-ACO highlights the superior performance of the heuristic formula described in Section 4.3. This formula enables ants to escape trap regions and efficiently find feasible paths, demonstrating the effectiveness of the improved heuristic adapted to the wide-area search strategy. Additionally, the data in Table 6 clearly show that PR-ACO outperforms other algorithms across various metrics. In contrast, other algorithms experience a significant increase in turning points and average runtime due to disorientation after entering trap areas. On this map, PR-ACO achieves an average runtime reduction of 31.8 % compared to D-ACO and 35 % compared to EH-ACO. In conclusion, the improved PR-ACO algorithm exhibits outstanding adaptability and performance in large-scale environments.

### 6.6. Physical experiment 1

To further validate the effectiveness and feasibility of the proposed algorithm, this section presents a physical experiment conducted in a complex environment. The physical verification is carried out using the Turoboat2 robotic platform, which is equipped with an industrial control computer, a Silan A2 LiDAR, and an XBOX depth camera. The control computer runs Ubuntu 16.04, with an integrated ROS system working in conjunction with the Turoboat2 robot,as shown in Fig 17. Physical Experiment 1 simulates a large corridor area with irregularly distributed goods, referred to in this study as Map1. The generated map covers a wide area of 602 × 602 grid units and includes multiple obstacles to mimic real-world complexities, as illustrated in Fig 18.

This chapter evaluates the effectiveness of the improved PR-ACO algorithm by assessing the quality of the paths it generates. To further ensure the validity of the comparative analysis, this chapter also uses other algorithms for comparison. The three algorithms are the proposed PR-ACO algorithm, a traditional eight-direction search algorithm (EH-ACO), and another improved algorithm (P1-ACO). The experiment was conducted with the same weight parameter settings as shown in Table 1 to provide a fair comparison.

During the experiment, it is first necessary to map the well-structured experimental area. By using the LiDAR to scan around the experimental region and applying the SLAM algorithm, a 2D planar map is generated, as shown in Fig 19. The white areas in the map represent traversable regions without obstacles, while the black areas indicate obstacle zones.

In practical applications, the traditional eight-directional search-based ACO algorithm performs extremely poorly, often failing to generate a valid path within the stipulated time, making it unsuitable as an effective comparison group in experiments. However, to demonstrate the effectiveness of the proposed algorithm, this paper selects the EH-

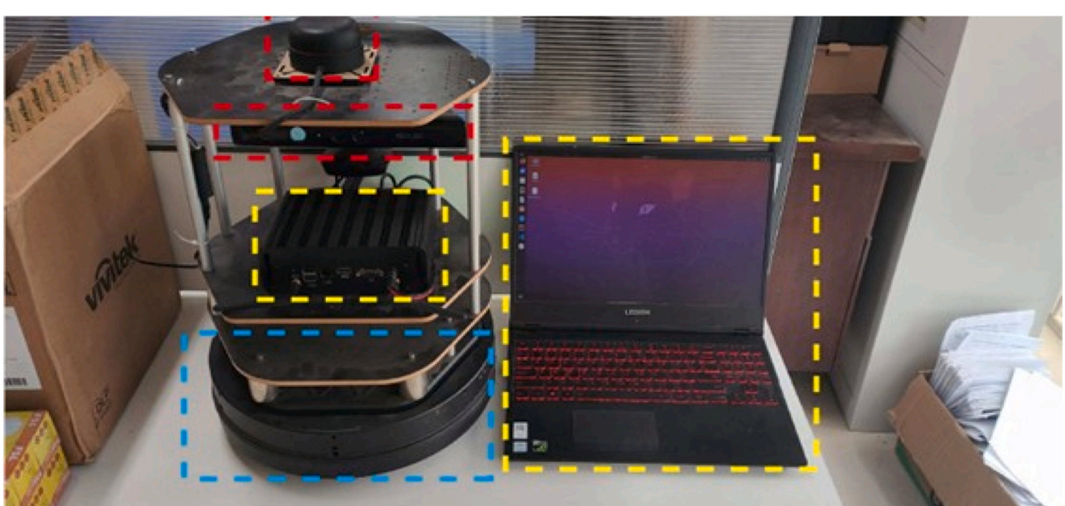


**Fig. 17.** The experimental mobile robot system, including the upper-level control unit (host computer).


**Fig. 18.** (a) and (b) Real-world experimental setup of Map1 from different perspectives.

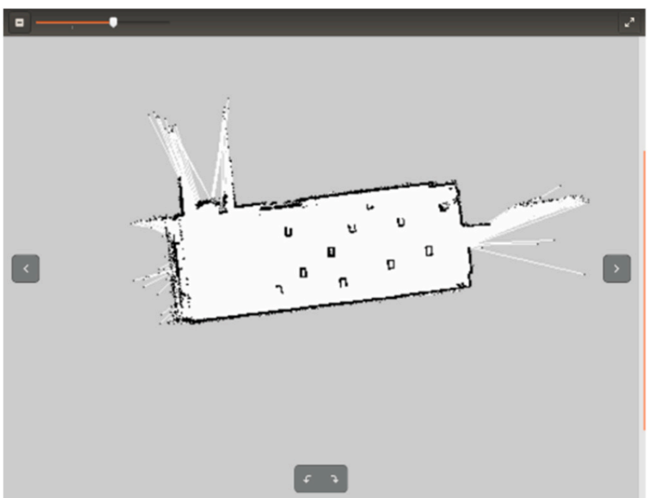


**Fig. 19.** The Map1 environment constructed by LiDAR for robot navigation. The white areas represent traversable, obstacle-free regions, while the black areas indicate obstacles.

ACO algorithm, which is also based on a traditional eight-directional search approach and performed relatively better in the comparative experiments conducted earlier, as the experimental benchmark.

To further validate the effectiveness of the additional improvements proposed in this paper, the P1-ACO algorithm is introduced. The P1-ACO algorithm adopts the wide-area search strategy proposed in this paper but retains some of the basic strategies mentioned in the EH-ACO algorithm, such as the heuristic formula and pheromone update principles. This allows for a direct comparison of the improvements introduced by this work.

As shown in Fig 20(a), the PR-ACO algorithm demonstrates excellent path-planning performance. It quickly generates a viable and efficient route, highlighting its strong adaptability. Moreover, the path produced by PR-ACO adheres well to practical robot motion constraints. In contrast, Fig 20(b) illustrates that while the EH-ACO algorithm is capable of producing a path, it tends to route too close to obstacles. This behavior results from its narrow search scope in large-scale maps, which causes it to behave similarly to a greedy algorithm—producing locally optimal paths that are difficult to adjust. Additionally, EH-ACO inherits the traditional ACO algorithm's drawback of excessive computation time, rendering it far less suitable for real-time applications compared to the other two algorithms.

In contrast, Fig 20(b) illustrates the performance of the EH-ACO algorithm, which, although capable of producing a path, generates routes that are excessively close to obstacles. This highlights the limitations of the EH-ACO algorithm, which, due to its overly narrow search scope in

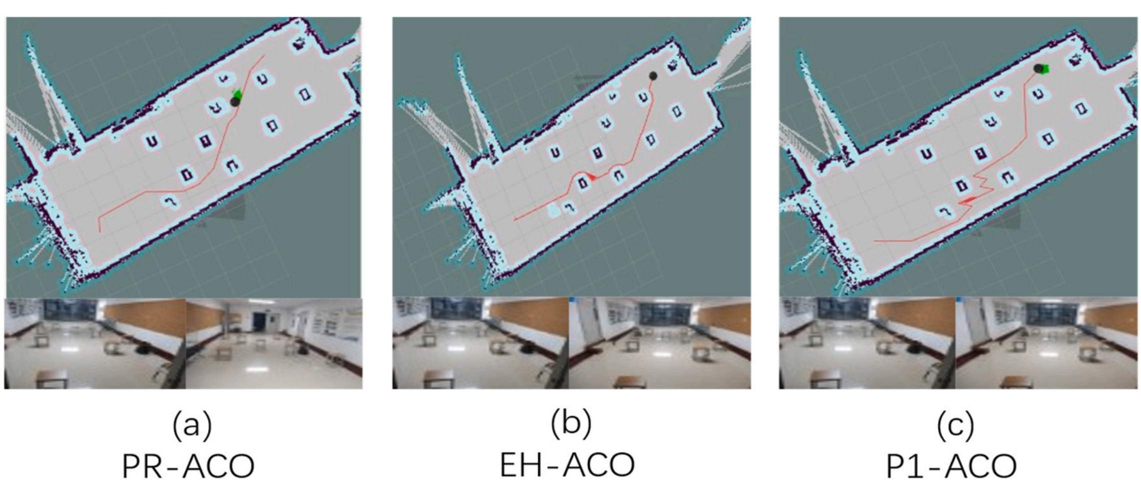



**Fig. 20.** The figure illustrates the motion trajectories generated during the physical experiments and the robot's actual movement directions.

**Table 7**
Average time of each algorithm over 10 runs on different maps.

| | Map1 | Map2 | Map1 | Map2 |
|---|---|---|---|---|
| | Average time | | Variance of time | |
| PR-ACO | 2.1736 | 2.5246 | 0.4661 | 0.4713 |
| EH-ACO | 29.5573 | 33.8406 | 7.6455 | 8.5489 |
| P1-ACO | 2.6942 | 3.1895 | 0.5319 | 0.9051 |

large-scale maps, ultimately exhibits a behavior similar to that of a greedy algorithm. The generated paths represent local optima and are difficult to adjust. Moreover, the EH-ACO algorithm inherits the shortcomings of traditional ACO algorithms, requiring significantly more time to generate paths. This leads to a severe loss of real-time performance, making it far less efficient than the other two algorithms.

Fig 20(c) shows that the P1-ACO algorithm benefits from the wide-area search strategy, achieving faster path generation and better timeliness. However, due to the lack of enhancements in heuristic adaptability, it exhibits some shortcomings in handling local environments effectively. As shown in Fig 20(a), the PR-ACO algorithm demonstrates excellent pathfinding performance. It quickly generates high-quality, practical paths within a short period, showcasing outstanding adaptability.

As shown in Table 7, the PR-ACO algorithm significantly outperforms the EH-ACO algorithm in terms of runtime on Map1, achieving a reduction of 92.7 %. Although the difference in running time between PR-ACO and P1-ACO is relatively small at 19.3 %, the path generated by PR-ACO is significantly smoother and more suitable for actual robot motion. This demonstrates the effectiveness of the improvements in the heuristic function and pheromone update strategy. In conclusion, the optimization of the PR-ACO algorithm is effective.

### 6.7. Physical experiment 2

To further validate the effectiveness of the proposed PR-ACO algorithm in real-world scenarios, this study designed an additional set of physical experiments in a different environment. Physical Experiment 2 was conducted in a small corridor space composed of two large halls connected by a narrow passage, filled with irregular obstacles. This environment is referred to as Map2 (see Fig. 21). The mobile robot platform and system configuration remained consistent with those used in Physical Experiment 1. Before testing, a detailed map of the environment was constructed using the SLAM algorithm. A 2D occupancy grid map was generated by navigating the robot around the perimeter using LiDAR. As shown in Fig. 22, In the resulting map, white regions indicate traversable space, while black areas represent obstacles. The final navigable space was determined by applying obstacle inflation layers.

To assess performance, PR-ACO was compared with EH-ACO and P1-ACO—the latter being an earlier variant that incorporates a global search strategy but lacks the improved heuristic enhancements. As shown in Fig 23(a), PR-ACO generated smoother and more reasonable paths. In contrast, Fig 23(b) illustrates that EH-ACO produced longer and less efficient paths in Map2, with a significantly increased runtime. Although P1-ACO, shown in Fig 23(c), benefited from faster runtimes due to its global search mechanism, it still fell short in terms of path quality and smoothness.

According to the data in Table VII, PR-ACO reduced average runtime by 95.8 % compared to EH-ACO, and by 17.7 % compared to P1-ACO, highlighting its superior computational efficiency. The time variance shows that PR-ACO has the smallest variance, enabling stable path updates and execution. In contrast, EH-ACO frequently fails to update or generates lost paths within the allotted time, showing very poor stability. P1-ACO, lacking some key PR-ACO improvements, also exhibits noticeable variance. These results confirm that the proposed PR-ACO algorithm delivers consistently strong performance across various

**Fig. 22.** The Map2 environment constructed by LiDAR for robot navigation. The white areas represent traversable, obstacle-free regions, while the black areas indicate obstacles.


**Fig. 21.** Figures (a), (b), and (c) show the real-world experimental setup of Map2 from different perspectives.

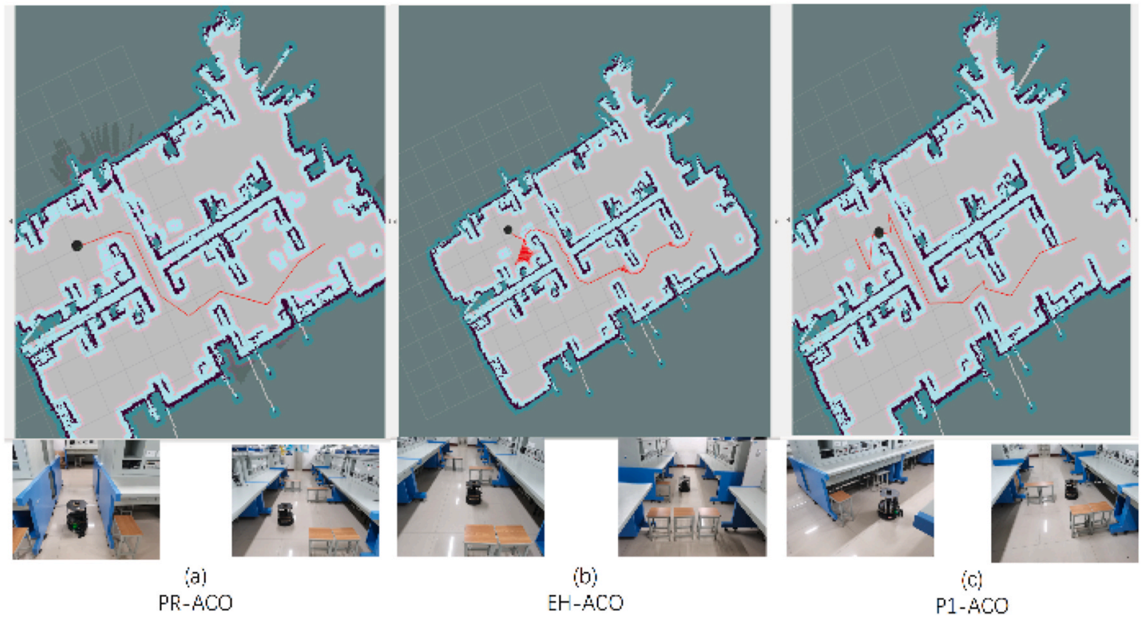



**Fig. 23.** The generated motion trajectory and the actual movement orientation of the robot in the physical experiment are shown.

environments, demonstrating excellent adaptability and global path planning capability. The effectiveness of the proposed optimizations is thus well supported.

## 7. Conclusion

To address the limitations of traditional Ant Colony Optimization (ACO) algorithms in mobile robot path planning, this paper proposes a novel algorithm—Probabilistic Roadmap integrated Ant Colony Optimization with a multi-strategy hybrid variable-region search mechanism, referred to as PR-ACO. The algorithm begins by modeling the environment using a 2D grid map and dynamically adjusts the search radius of ants based on map complexity, thereby enhancing coverage and completeness during a single iteration. Inspired by the Particle Swarm Optimization (PSO) paradigm, an auxiliary selection mechanism is introduced to exploit high-quality ants during the search process. A dynamic weighting coefficient is designed to improve heuristic information discrimination and search efficiency.

Furthermore, a pheromone topological diffusion mechanism is developed to enhance the utilization of pheromone information from previous iterations. To address convergence issues, a hierarchical decision-making strategy is employed based on elite ant principles. Elite paths are categorized into three types: optimal elite paths, abandoned paths, and sub-elite paths. These categories are used to assess the convergence quality of each iteration, supporting a dynamic pheromone update mechanism that avoids premature convergence and stagnation in local optima.

Both simulation and real-world experiments demonstrate the strong adaptability and performance of PR-ACO across various map scales. The proposed algorithm consistently finds high-quality and smooth paths, suitable for real-world robot motion constraints. It effectively overcomes challenges such as escaping trap regions and addressing local optimality. In physical experiments, PR-ACO not only successfully generates feasible paths from start to goal but also significantly reduces computation time compared to baseline methods, making it more efficient and practical for real-time robotic applications.

Looking forward, future work will aim to expand PR-ACO's applicability across both classical and quantum computing domains. In classical computing, focus will be placed on improving its scalability for large-scale and high-dimensional environments, refining region selection and pheromone mechanisms to further reduce computational overhead, and deploying the algorithm in real-world applications such as autonomous driving, warehouse logistics, and urban path planning. Integration with frameworks like ROS will enable seamless deployment on ground and aerial robotic platforms, including UAVs and autonomous vehicles.

In the realm of quantum computing, future studies will explore leveraging quantum parallelism and amplitude amplification to accelerate pathfinding and convergence, potentially enabling near-instantaneous optimization in highly complex or dynamic environments. This cross-domain adaptability positions PR-ACO as a promising candidate for next-generation intelligent navigation systems across terrestrial, aerial, and hybrid robotic ecosystems.

## CRediT authorship contribution statement

**Jianjuan Liu:** Supervision, Funding acquisition. **Yiheng Qian:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Data curation, Conceptualization. **Haibo Li:** Investigation. **Miaoxin Ji:** Funding acquisition. **Qiangwei Xu:** Funding acquisition. **Wenzhuo Zhang:** Investigation.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgements

Thank you for the support of the National Key R&D Program of China (2024YFC2208100), National College Students' Innovation and Entrepreneurship Training Program (Grant NO.202410463002), National Natural Science Foundation of China (Grant NO. 62201199), China Postdoctoral Science Foundation under Grant 2024M760841 and Henan University of Technology Self Science Innovation Fund Support Program (Grant NO.2021ZKCJ07). This work is supported by the Open Program of MOE Key Laboratory of Fundamental Physical Quantities Measurements National Province College Student Innovation Training Program (Grant NO.25B413003). We will also thank the reviewing experts for

their constructive feedback on our research.

## Appendix A. An example appendix

Authors including an appendix section should do so before References section. Multiple appendices should all have headings in the style used above. They will automatically be ordered A, B, C etc.

### A.1. Example of a sub-heading within an appendix

There is also the option to include a subheading within the Appendix if you wish.

## Data availability

Data will be made available on request.

## References


Bai, Xiangen, Liu, Di, Xu, L., Wang, J.Y., 2024. Cooperative path planning study of distributed multi-mobile robots based on optimized ACO algorithm. https://doi.org/10.1016/j.robot.2024.104748.

Butaney, S. (Sarah), Gaurav, K.Kumar, Ranjan, P. (Prabhat), Shrivas, N. V. (Nikhil Vivek), 2024. Recent developments in autonomous floor-cleaning robots: a review. Ind. Robot: Inter. J. Robot. Res. Appl. Early Access. https://doi.org/10.1108/IR-07-2024-0320. November.

Cai, Z., Liu, J.H., Xu, L., Wang, J.Y., 2024. Cooperative path planning study of distributed multi-mobile robots based on optimized ACO algorithm. https://doi.org/10.1016/j.robot.2024.104748.

Cao, X., Zou, X., Jia, C., Chen, M., Zeng, Z., 2019. RRT-based path planning for an intelligent litchi-picking manipulator. Comput. Electron. Agric. 156, 105–118. https://doi.org/10.1016/j.compag.2018.10.031.

Chen, T., Chen, S.F., Zhang, K.R., Qiu, G.T., Li, Q.P., Chen, X.M., 2022. A jump point search improved ant colony hybrid optimization algorithm for path planning of mobile robot. Int. J. Adv. Rob. Syst. 19 (5). https://doi.org/10.1177/17298806221127953.

Cheng, J., 2023. Dynamic Path Optimization Based on Improved Ant Colony Algorithm. https://doi.org/10.1155/2023/7651100.

Cheng, B., Wang, L., Tan, Q., Zhou, M., n.d. A deep reinforcement learning hyper-heuristic to solve order batching problem with mobile robots. https://doi.org/10.1007/s10489-024-05532-9.

Corominas, Guillem Rodríguez, Blesa, Maria J., Blum, Christian, 2023. AntNetAlign: ant colony optimization for network alignment. Appl. Soft Comput. 132, 109832. https://doi.org/10.1016/j.asoc.2022.109832. ISSN 1568-4946.

Fang, W.K., Liao, Z.G., Bai, Y.F., 2024. Improved ACO algorithm fused with improved Q-Learning algorithm for Bessel curve global path planning of search and rescue robots. https://doi.org/10.1016/j.robot.2024.104822.

Feng, Haobo, Hu, Qiao, Zhao, Zhenyi, Feng, Xinglong, 2024. Smooth path planning under maximum curvature constraints for autonomous underwater vehicles based on rapidly-exploring random tree star with B-spline curves. Eng. Appl. Artif. Intell. 133 (Part F), 108583. https://doi.org/10.1016/j.engappai.2024.108583. ISSN 0952-1976. https://www.sciencedirect.com/science/article/pii/S0952197624007413. Available at:

Fu, B., Chen, Y.M., Quan, Y., Zhou, X.L., Li, C.S., 2024. Bidirectional artificial potential field-based ant colony optimization for robot path planning. J. Robot. Autonomous Syst. 138, 103854. https://doi.org/10.1016/j.robot.2024.104834.

Gao, W., Tang, Q., Ye, B., Yang, Y., Yao, J., 2020. An enhanced heuristic ant colony optimization for mobile robot path planning. Soft Comput. 24, 6139–6150. https://doi.org/10.1007/s00500-020-04749-3.

Gao, Y., Wu, H., Wang, W., 2023. A hybrid ant colony optimization with fireworks algorithm to solve capacitated vehicle routing problem. Appl. Intell. 53, 7326–7342. https://doi.org/10.1007/s10489-022-03912-7.

Gong, D., Long, S., Dai, X., Zhang, Z., 2018. Mobile robot path planning based on ant colony Algorithm with A* heuristic method. Front. Neurorob. 13, 15. https://doi.org/10.3389/fnbot.2019.00015.

Guo, Jun, Xia, Wei, Hu, Xiaoxuan, Ma, Huawei, 2022. Feedback RRT* algorithm for UAV path planning in a hostile environment. Comput. Ind. Eng. 174, 108771. https://doi.org/10.1016/j.cie.2022.108771. ISSN 0360-8352.

Han, Z., Liu, S., Yu, F., 2017. A 3D measuring path planning strategy for intelligent CMMs based on an improved ant colony algorithm. Int. J. Adv. Manuf. Technol. 93, 1487–1497. https://doi.org/10.1007/s00170-017-0503-y.

Hu, Likun, Wei, Chunyou, Yin, Linfei, 2025. Fuzzy A* quantum multi-stage Q-learning artificial potential field for path planning of mobile robots. Eng. Appl. Artif. Intell. 141, 109866. https://doi.org/10.1016/j.engappai.2024.109866. ISSN 0952-1976.

Huang, Zhuo, Gao, Yang, Guo, Jian, Qian, Chen, Chen, Qingwei, 2024. An adaptive bidirectional quick optimal Rapidly-exploring Random Tree algorithm for path planning. Eng. Appl. Artif. Intell. 135, 108776. https://doi.org/10.1016/j.engappai.2024.108776. ISSN 0952-1976.

Jiao, Z.Q., Ma, K., Rong, Y.L., Wang, P., Zhang, H.K., 2018. A path planning method using adaptive polymorphic ant colony algorithm for smart wheelchairs. https://doi.org/10.1016/j.jocs.2018.02.004.

Kanna, B.R., Av, S.M., Hemalatha, C.S., Rajagopal, M.K., 2024. Enhancing SLAM efficiency: a comparative analysis of B-spline surface mapping and grid-based approaches. Appl. Intell. 54, 10802–10818. https://doi.org/10.1007/s10489-024-05776-5.

Lee, J., Seo, Y., 2024. Q-learning based on strategic artificial potential field for path planning enabling concealment and cover in ground battlefield environments. Appl. Intell. 54, 7170–7200. https://doi.org/10.1007/s10489-024-05436-8.

Li, T., Wang, C., Max Q.-H, M., Silva, C.W.d., 2019. Coverage sampling planner for UAV-enabled environmental exploration and field mapping. In: 2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), pp. 2509–2516. https://doi.org/10.1109/IROS40897.2019.8967735. Macau, China. https://ieeexplore.ieee.org/abstract/document/8967735.

Liang, Zuopeng, Yang, Yi, Huang, Zicheng, Zhou, Shibo, 2025. USV path planning algorithm based on 20D-A* and IDWA algorithms under COLREGs. Ocean. Eng. 331, 121389. https://doi.org/10.1016/j.oceaneng.2025.121389. ISSN 0029-8018.

Liu, Zhongpu, Liu, Jianjuan, 2023. Improved Ant Colony Algorithm for Path Planning of Mobile Robots Based on Compound Prediction Mechanism, pp. 2147–2162.

Liu, Y., Guo, S., Tang, S., Song, J., Zhang, J., 2025. Path planning for robots based on adaptive dual-layer ant colony optimization algorithm and adaptive dynamic window approach. IEEE Sens. J. 25 (11), 19694–19708. https://doi.org/10.1109/JSEN.2025.3557437, 1 June1.

Liu, J., Yang, J., Liu, H., et al., 2017. An improved ant colony algorithm for robot path planning. Soft Comput. 21, 5829–5839. https://doi.org/10.1007/s00500-016-2161-7.

Liu, X., Jiang, D., Tao, B., Jiang, G., Sun, Y., Kong, J., et al., 2022. Genetic algorithm-based trajectory optimization for digital twin robots. Front. Bioeng. Biotechnol. 9, 793782. https://doi.org/10.3389/fbioe.2021.793782.

Luo, Y., Guo, P., Zhang, M., 2019. A framework of ant colony P system. IEEE Access 7, 157655–157666. https://doi.org/10.1109/ACCESS.2019.2949952.

Luo, Y., Lin, S., Wang, Y., Liang, K., 2025. Path planning for mobile robots in complex environments based on enhanced sparrow search algorithm and dynamic window approach. Robotica 43 (5), 1929–1952. https://doi.org/10.1017/S026357472500061X.

Mao, P., Lv, S., Quan, Q., 2025. Tube RRT*: efficient homotopic path planning for swarm robotics passing-through large-scale obstacle environments. IEEE Rob. Autom. Lett. 10 (3), 2247–2254. https://doi.org/10.1109/LRA.2025.3531151.

Miao, X., Lee, J., Kang, B.-Y., 2018. Scalable coverage path planning for cleaning robots using rectangular map decomposition on large environments. IEEE Access 6, 38200–38215. https://doi.org/10.1109/ACCESS.2018.2853146.

Miao, C., Chen, G., Yan, C., Wu, Y., 2021. Path planning optimization of indoor mobile robot based on adaptive ant colony algorithm. https://doi.org/10.1016/j.cie.2021.107230.

Ravankar, A.A., Ravankar, A., Emaru, T., Kobayashi, Y., 2020. HPPRM: hybrid potential based probabilistic roadmap algorithm for improved dynamic path planning of Mobile robots. IEEE Access 8, 221743–221766. https://doi.org/10.1109/access.2020.3043333.

Russell, S., 2023. AI weapons: russia's war in Ukraine shows why the world must enact a ban. Nature 614 (7949), 620–623. https://doi.org/10.1038/d41586-023-00511-5.

Shi, K., Wu, W., Wu, Z., et al., 2024. Coverage path planning for cleaning robot based on improved simulated annealing algorithm and ant colony algorithm. SIViP 18, 3275–3284. https://doi.org/10.1007/s11760-023-02989-y.

Song, Baoye, Tang, Shumin, Li, Yao, 2024. A new path planning strategy integrating improved ACO and DWA algorithms for mobile robots in dynamic environments. Math. Biosci. Eng. 21 (2), 2189–2211. https://doi.org/10.3934/mbe.2024096.

Sui, Fuli, Tang, Xiaoke, Dong, Zihao, Gan, Xingjia, Luo, Peng, Sun, Jing, 2023. ACO+PSO+A*: a bi-layer hybrid algorithm for multi-task path planning of an AUV. Comput. Ind. Eng. 175, 108905. https://doi.org/10.1016/j.cie.2022.108905. ISSN 0360-8352.

Tan, Y., Ouyang, J., Zhang, Z., Lao, Y., Wen, P., 2023. Path planning for spot welding robots based on improved ant colony algorithm. Robotica 41, 926–938. https://doi.org/10.1017/S026357472200114X.

Tan, Y.S., Ouyang, J., Zhang, Z., Lao, Y.L., Wen, P.J., 2023. Path Planning for Spot Welding Robots Based on Improved Ant Colony Algorithm. https://doi.org/10.1017/S026357472200114X.

Tang, Y., Zakaria, M.A., Younas, M., 2025. Path planning trends for autonomous Mobile robot navigation: a review. Sensors 25 (4), 1206. https://doi.org/10.3390/s25041206.

Wang, W., Li, J., Bai, Z., Wei, Z., Peng, J., 2024. Toward optimization of AGV path planning: an RRT*-ACO algorithm. IEEE Access 12, 18387–18399. https://doi.org/10.1109/ACCESS.2024.3359748.

Wu, D., Du, K., 2025. A hybrid optimization algorithm of ACO and RRT for solving mobile robot path planning. Meas. Sci. Technol. 36, 076215. https://doi.org/10.1088/1361-6501/adee3a.

Wu, L., Huang, X., Cui, J., et al., 2023. Modified adaptive ant colony optimization algorithm and its application for solving path planning of mobile robot. Expert Syst. Appl. 215, 119410.

Wu, Q., You, X., Liu, S., 2024. Multi-ant colony optimization algorithm based on game strategy and hierarchical temporal memory model. Clust. Comput. 27, 3113–3133. https://doi.org/10.1007/s10586-023-04136-1.

Xia, G.Q., Han, Z.W., Zhao, B., Liu, C.Y., Wang, X.W., 2019. Global path planning for unmanned surface vehicle based on improved quantum ant colony algorithm. https://doi.org/10.1155/2019/2902170.

Yaacoub, J.P., Noura, H., Salman, O., Chehab, A., 2020. Security analysis of drones systems: attacks, limitations, and recommendations. Internet Things 11, 1–39. https://doi.org/10.1016/j.iot.2020.100218.

Yi, N., Xu, J., Yan, L., et al., 2020. Task optimization and scheduling of distributed cyber–physical system based on improved ant colony algorithm. Future Gener. Comput. Syst. 109, 134–148.

Yu, Miao, 2019. A solution of TSP based on the ant colony algorithm improved by particle swarm optimization. Discrete Continuous Dynam. Syst. - S 12 (4&5), 979–987. https://doi.org/10.3934/dcdss.2019066.

**Jianjuan Liu.** School of Electrical Engineering, Henan University of Technology, Zhengzhou, China.

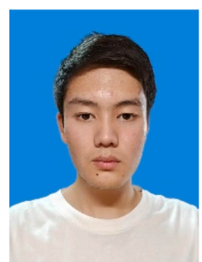
**Jianjuan Liu** received the Ph.D. degree in precision instrument and machinery from Southeast University, Nanjing, China, in 2007. She is currently a Professor with the School of Electrical Engineering, Henan University of Technology. Her main research interests include the nonlinear filtering and general estimation theory, and distributed signal processing for agent network navigation and position.

**Yiheng Qian** is currently pursuing a master's degree in Control Science and Engineering at Henan University of Technology in Zhengzhou, China. Obtained a Bachelor's degree in Building Electrical and Intelligent Engineering from North China University of Science and Technology in 2021. His current research interests include path planning for mobile robots and intelligent optimization algorithms. (Email: a1136501178@gmail.com)

**Li Haibo** (1999-), male, master candidate. His research interest is mobile robot path Planning Technology (E-mail) lhb176399799282022@163. com.

**Miaoxin Ji** received the B.E. degree from the Henan University of science and technology. Luoyang, China, in 2015. M.S. degree in mechanical engineering from Zhengzhou University, Zhengzhou, China, in 2018, and the Ph.D. degree in the School of Technology from the Beijing Forestry University, Beijing, China, in 2021. He is currently a lecturer and Master's Supervisor with the School of Electrical Engineering, Henan University of Technology, Henan, China. His research interests include pedestrian inertial navigation and pedestrian pose estimation. (Email: jimiaoxin2019@163.com)

**Qiangwei Xu** received Ph.D. degree in precision measurement physics from Huazhong University of Science and Technology, Wuhan, China, in 2023. He is currently a lecturer with the School of Electrical Engineering, Henan University of Technology. His research interests include the inertial sensors, capacitive transducers, dynamic range of the accelerometer and weak signal detection.



**Wenzhuo Zhang.** School of Electrical Engineering at Henan University of Technology, Zhengzhou, China, majors in Automation. He participated in the "Xinyingda Cup" Embedded Innovation Design Competition organized by the Henan Instrument and Meter Association and won the first prize.